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Chapter 1: Introduction of Power query and Clean Raw Data

What is Power Query?

Power Query is a tool inside Excel (and also in Power BI) that helps you **import, clean, and transform data automatically**—without doing everything manually.

Think of Power Query as a “**Data Cleaning Machine**” inside Excel.

You give it your messy data, and it cleans, arranges, and prepares it the way you want—**just once**. After that, you can **refresh** it anytime, and it will do all the cleaning again automatically.

Why is Power Query Useful for Excel Users?

Most Excel users face these problems:

- Data comes from different sources (CSV, Excel files, websites, text files).
- Data is messy—wrong spelling, blank rows, extra spaces, wrong formats.
- You spend hours cleaning the same type of data again and again.
- Merging multiple sheets/files manually takes a lot of time.

👉 **Power Query solves all these problems** by automating the cleaning and combining steps.

A Simple Layman Example

Imagine you run a restaurant and every week you get a customer list in Excel like this:

- Some names are in CAPITAL
- Some rows are blank
- Some phone numbers have dashes, some don't
- Some duplicate entries exist

Normally: You will clean this manually every time.

With Power Query:

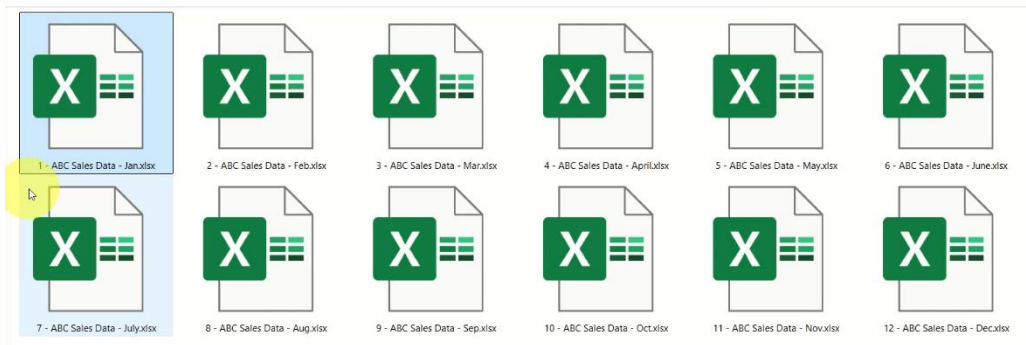
- Remove blank rows
- Remove duplicates
- Fix text case
- Standardize phone numbers

And the best part?

Next week, you just click **Refresh** and it cleans everything again automatically!

Example 2: Combining Sales Files

You receive 12 monthly sales files:
Jan.xlsx, Feb.xlsx, ... till Dec.xlsx



Without Power Query:

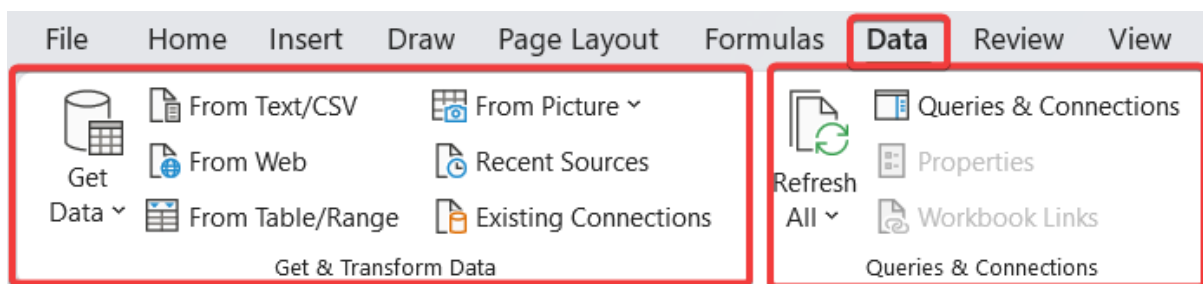
- You open each file
- Copy-paste into one sheet
- Remove blank columns/rows
- Fix formatting

With Power Query:

- Select folder → combine all files in one go
- Clean once → Click refresh next time

Where to Find Power Query in Excel?

📌 **Data Tab → Get & Transform Data**



Or in Old Excel versions:

📌 **Power Query Tab** (Excel 2010/2013 add-in) – But you need to install it separately

What Can Power Query Do?

Here are some powerful things it can handle easily:

Import data from:

- Excel files
- CSV files
- Folders
- SQL servers
- Websites
- Text files

Clean and transform data:

- Remove duplicates
- Remove blank rows
- Split columns
- Combine tables
- Change data type
- Replace values
- Group data

Automate repeated tasks:

- Once set up, just **refresh** instead of doing the same work again

Why Data Users Love Power Query

1. Saves Time

What used to take 2 hours can now be done in 2 minutes.

2. No Coding Needed

You don't need to be a programmer. All actions are menu-based.

3. No Manual Rework

Just refresh to apply the same steps again.

4. Handles Big Data Easily

Unlike normal Excel sheets, Power Query can process large data smoothly.

5. Less Chance of Mistakes

Automated steps mean less human error.

Basic Workflow of Power Query

1 Import the Data

(From Excel, CSV, folder, website, etc.)

2 Clean the Data

(Remove errors, blanks, fix format)

3 Transform the Data

(Split columns, filter, merge, rename)

4 Load the Data

(Into Excel sheet or Power Pivot)

5 Refresh When Needed

(Everything updates automatically)

We Will Start with Real-Life Example of Cleaning Data

ID	Date of collection date	Age	Gender	Name Initial	Last Name	Hypertension	Heart Disease	Ever Married	Work Type	Residence Type	Average Glucose Level	BMI
1	04-20-2024	78	Female	A	Roy	0	0	1	Private	Rural	267.3	36.2
2	04-20-2024	60	Female	A	Shenoy	0	0	0	Private	Rural	207.24	19.8
3	04-20-2024	69	Male	S	Geloy	0	0	0	Private	Urban	161.3	34.5
4	04-20-2024	43	Male	G	Messy	0	0	1	Never Worked	Urban	247.49	48.8
5	04-21-2024	30	Female	B	Bern	0	0	1	Government	Urban	116.57	31.3
6	04-22-2024	53	M	S	Bush	1	0	1	Government	Urban	220.62	49.9
7	04-22-2024	83	Female	W	Bern	0	0	0	Children	Urban	193.32	46.8
8	04-22-2024	43	Female	N	Jordan	0	0	0	Never Worked	Urban	198.59	23.3
9	04-22-2024	62	Male	M	Jordan	1	0	0	Children	Rural	117.81	16.5
10	04-22-2024	35	Male	A	Sundar	0	0	0	Government	Urban	289.09	37
11	04-26-2024	83	M	L	Armstrong	0	0	1	Private	Urban	150.86	34.3
12	04-27-2024	71	Male	M	Fritzler	0	0	1	Private	Urban	254.9	49.5
13	04-28-2024	79	Male	K	Black	0	0	0	Self-employed	Rural	164.44	11.4
14	04-29-2024	50	Male	J	Bell-	0	0	1	Self-employed	Rural	142.49	15.2
15	04-30-2024	25	M	D	Eno	0	0	1	Government	Rural	218.1	36.5
16	04-30-2024	81	F	D	percer	0	0	0	Private	Rural	245.06	10.9
17	04-30-2024	80	Male	R	Collins	0	1	1	Private	Urban	266.21	18.9
18	04-30-2024	45	F	W	Brown	0	0	1	Self-employed	Rural	221.45	34.5
19	04-30-2024	47	Male	B	Eplett	0	0	1	Private	Rural	131.46	24

Image 1

Example: Cleaning Patient Medical History Data with Power Query

In the **Image 1 Screenshot**, you can see a table that contains patient medical history details. This data is **raw** and not ready for analysis. Before we can use it, we need to clean and standardize it properly.

If we try to clean all this directly in Excel, it will take time and effort. And the biggest problem is — if new rows are added tomorrow, we will have to repeat all the steps again manually.

👉 **That's why we will clean this data using Power Query — so that the process becomes automatic and refreshable.**

Action

Select data – Click on Data Tab – Click on From Table or Range – Power query window will appear

	P	Q	R	S	T	U	V	W	X	Y	Z	AA
81	Non-Vegetarian	0	0	7	0	Secondary	Middle	37	North	0		
82	Non-Vegetarian	0	0	12	0	Tertiary	Low	9	East	0		
83	Non-Vegetarian	0	0	10	0	Secondary	Middle	98	West	0		
84	Vegetarian	0	0	3	0	Secondary	Middle	95	North	0		
85	Vegetarian	0	1	10	1	Primary	Middle	68	North	0		

Cleaning Tasks We Need to Perform

Here are the main tasks that need to be done before analysis:

1 Remove Blank Rows

There are empty records in the data that don't contain any useful information. These blank rows can create wrong outputs in reports, charts, and analysis. So we must delete them.

Power Query Step:

- Go to **Home** → **Remove Rows** → **Remove Blank Rows**
-

2 Change the Date Format to DD-MM-YYYY

The date column might be in different formats like MM/DD/YYYY or YYYY-MM-DD. For consistency and proper sorting, we will convert it to **DD-MM-YYYY**.

Power Query Step:

- Select the Date column
 - Go to **Transform** → **Data Type** → **Date**
 - Right-click the column → **Change Type using Locale**
 - Select **English (India)** or a locale that uses DD-MM-YYYY
-

3 Replace Gender Codes (M/F) with Full Words

Right now, the Gender column contains short codes like **M** and **F**. For better understanding and proper reporting, we will change them to **Male** and **Female**.

Power Query Step:

- Select the Gender column
 - Go to **Transform** → **Replace Values**
 - Replace M with Male
 - Replace F with Female
-

4 Combine Two Name Columns (Initial + Last Name)

There are two separate columns, for example:

- Column E: Initial or First Name
- Column F: Last Name

We need to merge them into one full name column.

Power Query Step:

- Select both columns (E and F)
- Go to **Transform → Merge Columns**
- Choose a **space** as the separator
- Name the new column: **Full Name**

5 Convert Marital Status into Codes (0 & 1)

Let's say the Marital Status column contains words like **Married** and **Unmarried**. We want to convert them into numeric codes for analysis:

- **0 = Unmarried**
- **1 = Married**

Power Query Step:

- Select the Marital Status column
- Go to **Transform → Replace Values**
 - Replace Unmarried with 0
 - Replace Married with 1
- Change the column type to **Whole Number**
→ Right-click column → **Change Type → Whole Number**

Why Do This in Power Query Instead of Excel?

If you clean this in Excel manually, every time someone adds new data, you must repeat all steps again (remove blanks, combine names, change gender, etc.).

But in **Power Query**, you just set it up **once** and use **Refresh** next time. All the same steps will be applied automatically on new or updated data.

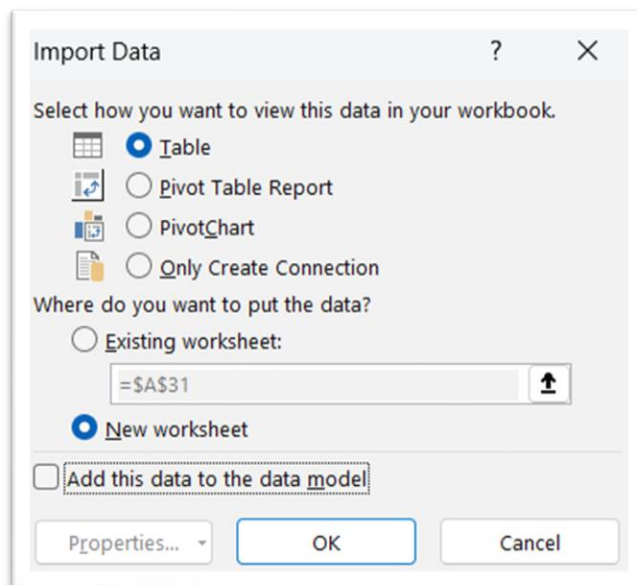
Summary Table of Cleaning Tasks & Power Query Actions

Task	Problem	Power Query Solution
1	Blank rows	Home → Remove Rows → Remove Blank Rows
2	Wrong date format	Transform → Data Type → Change Type using Locale
3	Gender codes (M/F)	Transform → Replace Values (M → Male, F → Female)
4	Name split in two columns	Transform → Merge Columns (Space as separator)

5	Marital Status text	Replace Values (Unmarried → 0, Married → 1) → Change Type to Number
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Last Action – Once Data cleaning done – Click on Home Tab Close and Load To – which helps you to load data in Table, Pivot Table, Pivot Charts and Queries Connections.

Home Tab – Close and Load To



Chapter 2: Importing Data from a Website using Power Query

❖ Step-by-Step Process:

1. Visit the Website:

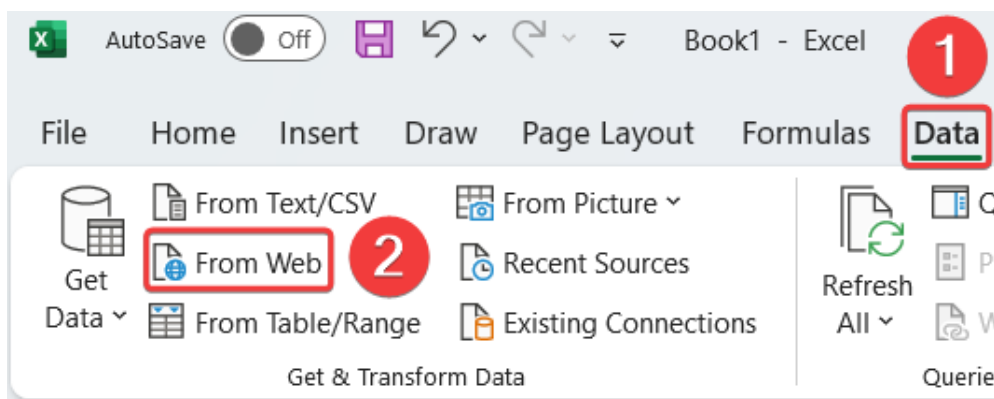
- Open your web browser and go to the website from which you want to import data (e.g., Worldometer).
- Copy the **URL** from the address bar.

2. Open Excel Workbook:

- Launch Microsoft Excel.
- Open a **blank workbook** or an existing one.

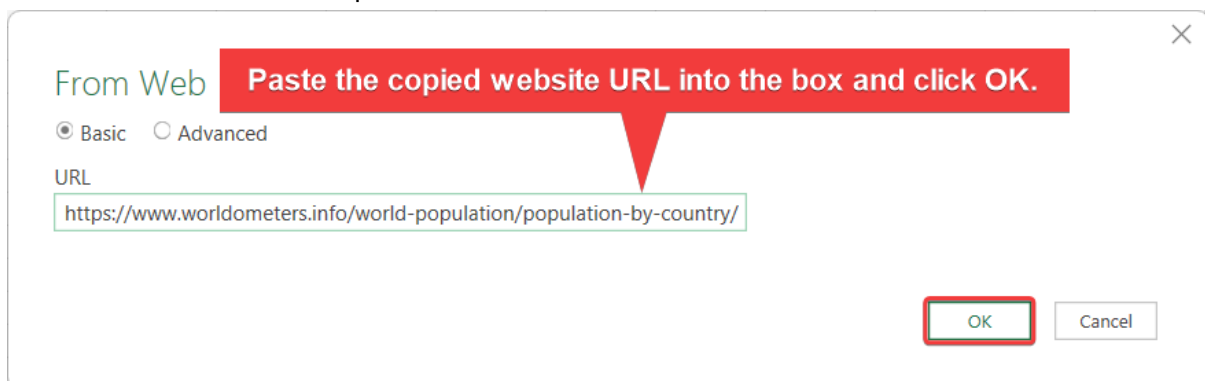
3. Navigate to the Data Tab:

- Click on the **Data** tab in the ribbon.
- Select **From Web** under the **Get & Transform Data** section.



4. Paste the URL:

- A dialog box will appear.
- Website URL: -https://www.worldometers.info/world-population/population-by-country/
- Paste the copied website URL into the box and click **OK**.



5. Load Data through Power Query:

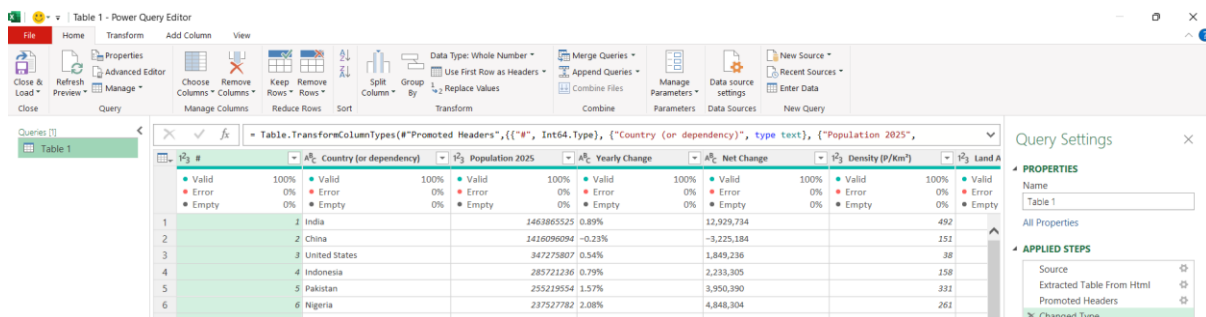
- Power Query will connect to the website and display a **Navigator Pane** with all available tables on the web page.
- You will see a **preview** of each table.

6. Select Table(s):

- Choose the relevant **table** (or multiple tables) you want to import.
- Click on **Transform Data** if you want to modify the data before importing.

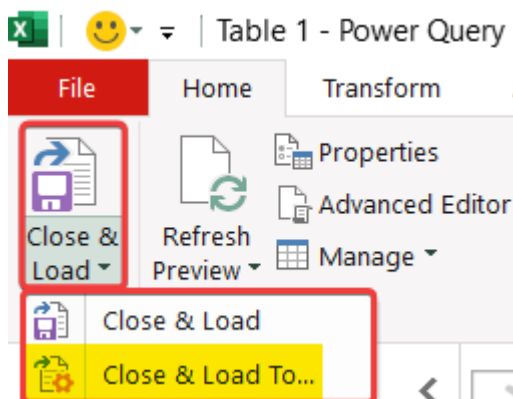
7. Transform the Data (Optional but Recommended):

- Use Power Query Editor to:
 - Remove blank rows
 - Remove duplicate entries
 - Rename columns
 - Change data types
 - Filter rows
 - Clean the data as needed

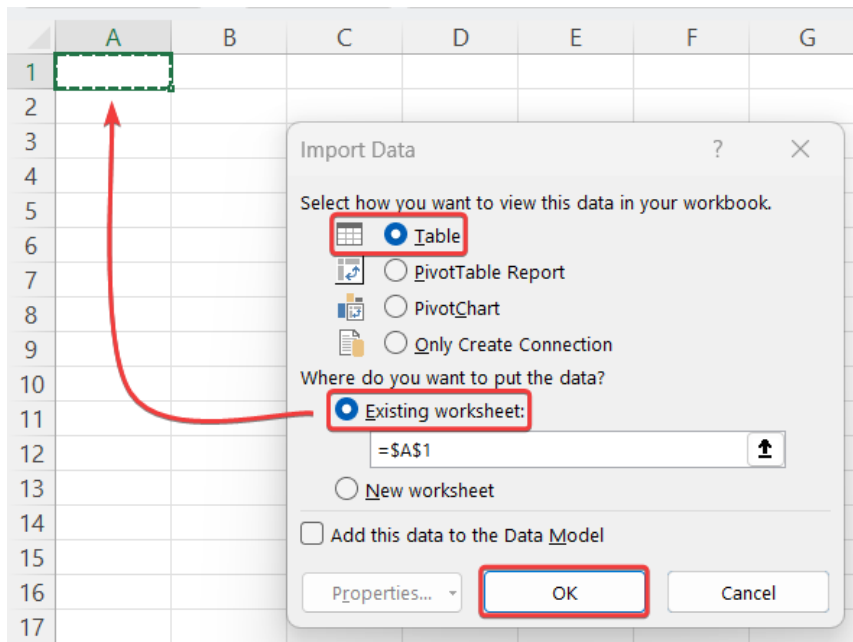


8. Load the Data:

- Once satisfied with the transformation, click on:
 - Close & Load** – to load the data into Excel.
- You can also rename the query (e.g., **Population_2025**) from the query settings.



- Last Action** – Once Data cleaning done – Click on **Home Tab Close and Load To** – which helps you to load data in Table, Pivot Table, Pivot Charts and Queries Connections.



#	Country (or dependency)	Population 2025	Yearly Change	Net Change	Density (P/Km ²)	Land Area (Km ²)	Migrants (net)	Fert. Rate	Median Age	Urban Pop %
1	India	1463865525	0.89%	12,929,734	492	2973190	-495,753	1.94	28.8	0.371
2	China	1416090994	-0.23%	-3,225,184	151	9388211	-268,126	1.02	40.1	0.675
3	United States	347276807	0.54%	1,849,236	38	9147420	1,230,663	1.62	38.5	0.828
4	Indonesia	285721236	0.79%	2,233,305	158	1811570	-39,509	2.1	30.4	0.596
5	Pakistan	255219554	1.57%	3,950,390	331	770880	-1,235,336	3.5	20.6	0.344
6	Nigeria	237527782	2.08%	4,848,304	261	910770	-15,258	4.3	18.1	0.549
7	Brazil	212812405	0.38%	813,832	25	8358140	-217,283	1.6	34.8	0.914

Key Features and Benefits:

Feature	Description
Live Data Refresh	The imported data can be refreshed automatically when the source website updates.
Web Table Detection	Power Query automatically detects HTML tables from web pages.
Data Cleaning Tools	Provides tools for data transformation like removing nulls, changing types, filtering, etc.
Reusable Queries	Queries can be saved and reused whenever needed.

Tips:

- Not all websites support structured table import. Prefer websites that display data in **HTML tables**.
- If the page uses JavaScript to load data, Power Query might **not detect** those tables.
- You can schedule data refresh or **manually refresh** via Data > Refresh All.

Step-by-Step: Importing Data from a PDF

❖ Step-by-Step Process:

1. Open the PDF File:

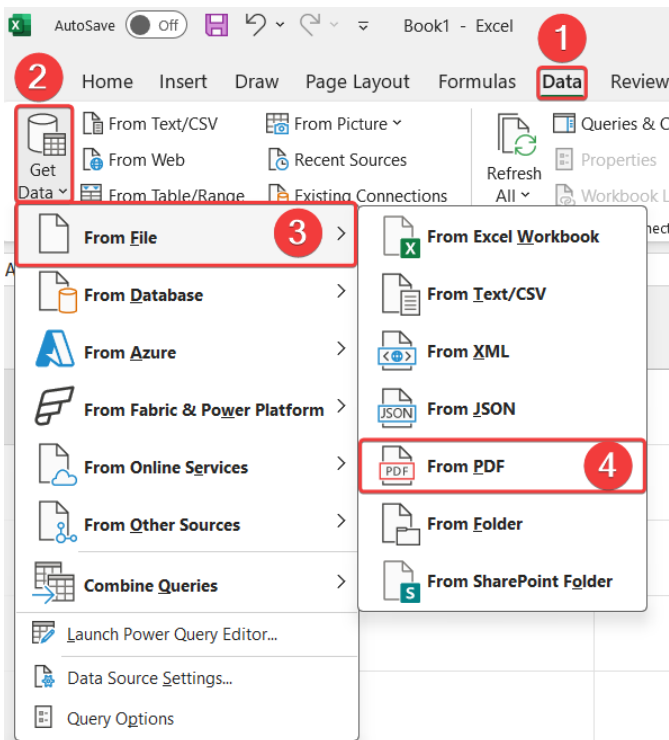
- Make sure you have the PDF file with the data (tables) you want to import.
- Keep it saved locally on your computer.

2. Open Excel Workbook:

- Launch Microsoft Excel and open a **blank or existing workbook**.

3. Go to the Data Tab:

- Click on the **Data** tab in the Excel ribbon.
- Select **From PDF** under the **Get & Transform Data** group.



4. Select the PDF File:

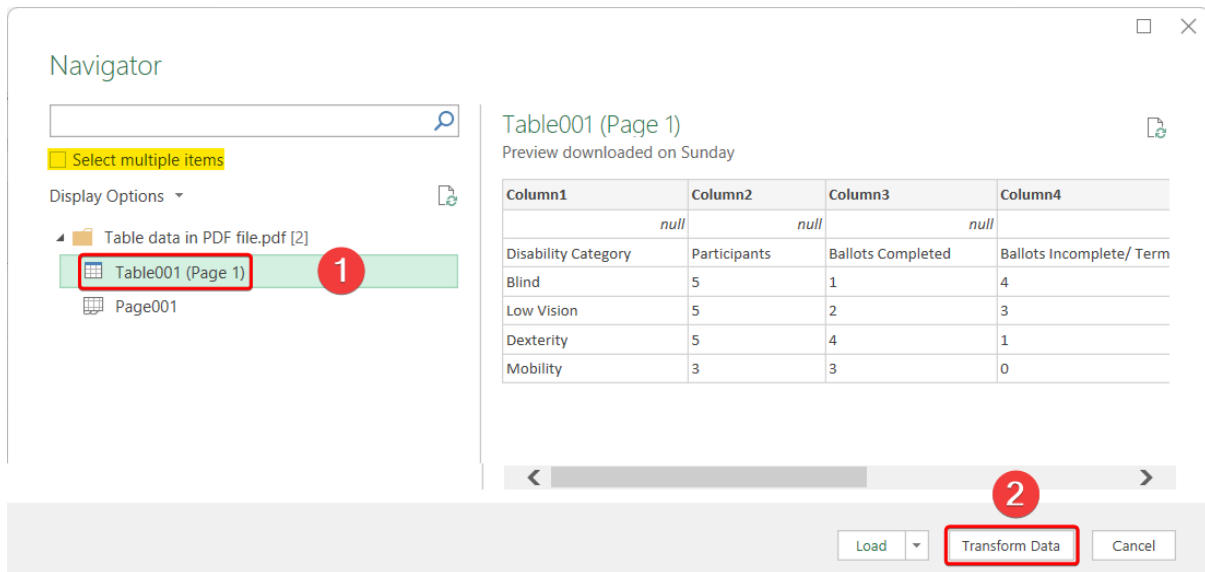
- Browse your computer to locate the PDF file.
- Select the file and click **Import**.

5. Navigator Preview Window:

- Power Query will scan the PDF and display a **list of tables and pages** it detects.
- Select the **table(s)** you want to extract. You can click on each to preview it.

6. Clean and Transform Data (Optional):

- Click on **Transform Data** to open the Power Query Editor.



- Use the editor to:
 - Remove blank or irrelevant rows
 - Promote the first row to headers
 - Rename columns
 - Change data types
 - Filter out unnecessary data

7. Load Data to Excel:

- Once you're happy with the cleaned data, click:
 - **Close & Load** to bring the data into Excel as a table.

Key Features and Benefits:

Feature	Description
Automatic Table Detection	Power Query scans PDF files and automatically identifies structured tables .
Multi-table Support	You can import multiple tables from a single PDF file.
Built-in Cleaning Tools	Easily clean and shape your data before loading it into Excel.
Repeatable Workflow	You can refresh or modify the query later without starting over.

Step-by-Step: Importing Data from an Image

❖ Step-by-Step Process:

1. **Prepare the Image File:**

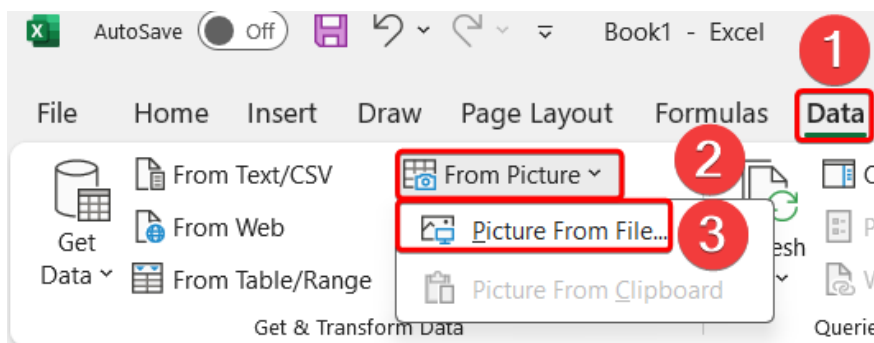
- Make sure your image contains a **clear and structured table**.
- Save the image in a supported format like .jpg, .png, or .bmp.

2. **Open Excel Workbook:**

- Start Excel and open a **new or existing** workbook.

3. **Go to the Data Tab:**

- In the ribbon, click on the **Data** tab.
- Choose **From Picture > Picture From File**.

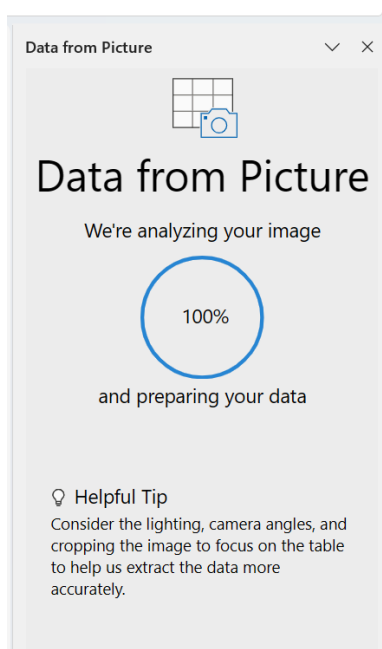


4. **Select the Image:**

- A file dialog will open.
- Locate and select the image file you want to import.

5. **OCR Processing:**

- Excel will use **OCR** to analyse and convert the image into text-based table data.
- A preview pane will appear with the **extracted data**.



6. **Review and Edit the Data:**

- You can make corrections if OCR misread any values (common in blurry or handwritten text).
- Make sure column headers and row values are correctly aligned.

Data from Picture

	Total defects	A	B	C	D	E
A4636	131	37	21	28		45
A2524	86	20	24	21	1	20
A3713	75	17	13	18		27
A4452	73	5	33	17		18
A4088	72	14	16	12	2	28
A2103	68	14	13	14	1	26
A2156	68	16	13	19	2	18
A3681	66	12	16	9	1	28
A1366	50	11	15	12		12
A2610	39	5	7	12		15
Total	728	151	171	162	7	237

click Insert to load it into the Excel sheet.

Insert Data

Review

	Total defects	A	B	c	D	E
A4636	131	37	21	28		45
A2524	86	20	24	21	1	20
A3713	75	17	13	18		27
A4452	73	5	33	17		18

	A	B	C	D	E	F	G
1		Total defects	A	B	c	D	E
2	A4636	131	37	21	28		45
3	A2524	86	20	24	21	1	20
4	A3713	75	17	13	18		27
5	A4452	73	5	33	17		18
6	A4088	72	14	16	12	2	28
7	A2103	68	14	13	14	1	26
8	A2156	68	16	13	19	2	18
9	A3681	66	12	16	9	1	28
10	A1366	50	11	15	12		12
11	A2610	39	5	7	12		15
12	Total	728	151	171	162	7	237

7. Insert Data into Excel:

- Once you're satisfied with the data, click **Insert** to load it into the Excel sheet.

Key Features and Benefits:

Feature	Description
OCR Technology	Uses Optical Character Recognition to convert images into editable table data.
Editable Preview	Allows you to review and edit data before inserting it into Excel.
Flexible Image Sources	Works with screenshots, scanned paper tables, photos, and more.
Quick Conversion	Saves time compared to manual typing from paper or image-based sources.

What is Pivoting and Unpivoting?

In Power Query, **Pivoting** and **Unpivoting** are powerful transformation tools used to **reshape your data**:

- **Pivoting:** Converts rows into columns.
- **Unpivoting:** Converts columns into rows.

These tools are especially helpful when preparing data for analysis, charts, or reports.

Step-by-Step: Pivoting and Unpivoting in Power Query

❖ Unpivoting Data (Columns → Rows)

1. Load Your Data into Power Query:

- Select your data and go to **Data > From Table/Range** to load it into Power Query.

The screenshot illustrates the initial step of loading data into Power Query. The 'Data' tab is selected, and the 'From Table/Range' option is highlighted. A table with 4 columns (Product, Category, Attribute, Value) is selected in the worksheet. A 'Create Table' dialog box is open, asking 'Where is the data for your table?' with the range '\$A\$1:\$D\$46' entered. The 'My table has headers' checkbox is checked. The table data is visible in the background, showing products like Soap, A2 Battery, Oil, and Baby Powder across different zones and their values.

2. Select the Data Range

- Click anywhere inside the data range, or
- Press **Ctrl+A** to select the entire data.

3. Go to the Data Tab

- Click on the **"Data"** tab from the Excel ribbon.

4. Choose 'From Table/Range'

- Click on **"From Table/Range"** option under the **Get & Transform Data** group.

5. Create Table Dialog Box

- A dialog box will appear.
- Ensure the selected range is correct.
- Check the box **"My table has headers"** if your data includes headers.
- Click **OK** to proceed.

The screenshot shows the Power Query Editor interface. The **Transform** tab is selected, and the **Pivot Column** button is highlighted with a red circle (3). Below the ribbon, the data table is visible with columns: Product, Category, Attribute, and Value. The **Attribute** column is selected with a red circle (1). The **Pivot Column** dialog box is open, showing the **Values Column** dropdown set to **Value** (4). The **OK** button is highlighted with a red circle (5).

	Product	Category	Attribute	Value
1	Soap	Household	Zone1	17
2	Soap	Household	Zone2	50
3	Soap	Household	Zone3	29

6. Rename Columns (Optional):

- Rename **"Attribute"** and **"Value"** columns to something meaningful (e.g., **"Year"** and **"Sales"**).

❖ Pivoting Data (Rows → Columns)

1. Load Your Data into Power Query:

- Ensure your data has at least:
 - A **category column** (e.g., Year)
 - A **values column** (e.g., Sales)

2. **Select the Column to Pivot:**

- Select the **column with repeating values** you want to turn into column headers (e.g., "Year").

3. **Click Pivot Column:**

- Go to **Transform > Pivot Column**.

4. **Choose Values Column:**

- In the dialog box, select the **Values Column** (e.g., "Sales") that will fill the pivoted cells.

5. **Click OK to Apply:**

- Your selected row values are now converted into **new columns**.

Key Features and Benefits:

Feature	Description
Pivoting	Converts row values into columns (great for creating summary tables).
Unpivoting	Converts columns into rows (great for flattening wide datasets).
Restructure Data Easily	Reshape messy or poorly formatted data into analysis-friendly formats.
Supports BI Tools	Ideal for preparing data for PivotTables , Power BI , and charts.

Step-by-Step: Appending Data from Multiple Sheets

❖ **Step-by-Step Process:**

1. **Convert Each Sheet into a Table:**

- Go to each sheet in your workbook.
- Select the data and press **Ctrl + T** to convert it into a table.
- Make sure headers are checked.

2. **Rename Each Table:**

- Give each table a meaningful name like:
 - Table_2022
 - Table_2023
 - Table_2024, etc.
- Use the **Table Design tab** to rename.

1. Table Design tab is selected in the ribbon.

2. Table Name: Table_2022 is highlighted in the Name box.

Date	Sales Rep	Customer	Amount	Profit	Region
01-Jan-22	Watson	Namint Enterprises	12850	5380	North
09-Jan-22	D'Souza	VCC	14400	5340	East
10-Jan-22	Roma	Jindle Power Works	14700	4660	South
14-Jan-22	Mark	MNTL	13350	5600	North
16-Jan-22	Steve	VCC	13850	4460	West
21-Jan-22	Vector	Jindle Power Works	11200	3420	South
21-Jan-22	Veronica	Data Tronics	13500	3660	North
24-Jan-22	Watson	Namint Enterprises	13350	3520	East
27-Jan-22	D'Souza	White Associates	14500	3580	East
29-Jan-22	Roma	VCC	14500	4180	East
30-Jan-22	Mark	Namint Enterprises	13150	2400	East
01-Feb-22	Steve	Jindle Power Works	12250	5280	West
01-Feb-22	Vector	White Associates	14150	5120	West
03-Feb-22	Veronica	Shah Associates	13600	3580	North
04-Feb-22	Ramesh	VCC	12300	2560	West
06-Feb-22	Bruce	Namint Enterprises	11950	3320	South
06-Feb-22	Varsha	Good Fly	11050	3740	North
14-Feb-22	Mark	Shyam & Sharma Co	11500	3220	East
14-Feb-22	Rajat	Data Tronics	11450	3800	East
17-Feb-22	Veronica	Shyam & Sharma Co	11050	3120	West
17-Feb-22	James	India Trotters	10800	2660	West
21-Feb-22	Anshika	Sharma & Co	12400	4980	East

3. Open Power Query Editor:

- Go to the **Data** tab.
- Click on **Get Data > Launch Power Query Editor**.

1. Data tab is selected in the ribbon.

2. Get Data button is highlighted in the ribbon.

3. Launch Power Query Editor... option is selected in the dropdown menu.

4. Fetch All Tables in Workbook:

- In the formula bar, enter:
- = Excel.CurrentWorkbook()
- This will return a list of **all named tables** in the current workbook.

The screenshot shows the Power Query Editor interface. The formula bar contains the text `= Excel.CurrentWorkbook()`. Below the formula bar, a table displays the results of the query, listing named tables in the workbook. The table has columns for 'Content' and 'Name'. The 'Content' column shows validation status (Valid, Error, Empty) and percentage. The 'Name' column lists the table names: Table_2022, Table_2023, and Table_2024. Red circles 1 through 5 highlight specific UI elements: 1. Home tab, 2. New Source dropdown, 3. Other Sources, 4. Blank Query, 5. Formula bar.

5. Select and Filter Required Tables:

- In the preview, identify the tables you want to append.
- You can filter by table name or remove any you don't need.

ABC 123	Content	AB C	Name
	Valid 100%	Valid 100%	
	Error 0%	Error 0%	
	Empty 0%	Empty 0%	
1	Table		Table_2022
2	Table		Table_2023
3	Table		Table_2024

Date	Sales Rep	Customer	Amount	Profit	Region
01-Jan-2022 12:00:00 AM	Watson	Namint Enterprises	12850	5380	North
09-Jan-2022 12:00:00 AM	D'Souza	VCC	14400	5340	East
10-Jan-2022 12:00:00 AM	Roma	Jindle Power Works	14700	4660	South
14-Jan-2022 12:00:00 AM	Mark	MNTL	13350	5600	North
16-Jan-2022 12:00:00 AM	Steve	VCC	13850	4460	West
21-Jan-2022 12:00:00 AM	Vector	Jindle Power Works	11200	3420	South
21-Jan-2022 12:00:00 AM	Veronica	Data Tronics	13500	3660	North
24-Jan-2022 12:00:00 AM	Watson	Namint Enterprises	13350	3520	East
27-Jan-2022 12:00:00 AM	D'Souza	White Associates	14500	3580	East
29-Jan-2022 12:00:00 AM	Roma	VCC	14500	4180	East
30-Jan-2022 12:00:00 AM	Mark	Namint Enterprises	13150	2400	East
01-Feb-2022 12:00:00 AM	Steve	Jindle Power Works	12250	5280	West
01-Feb-2022 12:00:00 AM	Vector	White Associates	14150	5120	West
03-Feb-2022 12:00:00 AM	Veronica	Shah Associates	13600	3580	North
04-Feb-2022 12:00:00 AM	Ramesh	VCC	12300	2560	West
06-Feb-2022 12:00:00 AM	Bruce	Namint Enterprises	11950	3320	South
06-Feb-2022 12:00:00 AM	Varsha	Good Fly	11050	3740	North
14-Feb-2022 12:00:00 AM	Mark	Shyam & Sharma Co	11500	3220	East

How to View Table Data in Power Query:

6. Do Not Click on the Table Name

- Avoid clicking directly on the **Table Name** in the column (like Table, Table1, etc.).

7. Click on the Blank Space (Cell beside the Table Name)

- Instead, click on the **blank space** or **cell beside the Table** entry (just to the right of the word "Table").

8. View the Table Preview

- Once you click the blank space, a **preview of the table** will appear at the bottom of the Power Query window.

Step-by-Step Guide

9. Locate the Expand Icon

- Look for this icon:
- It appears next to columns that contain records or tables.
- **Important:** This icon is called the **Expand Button**.

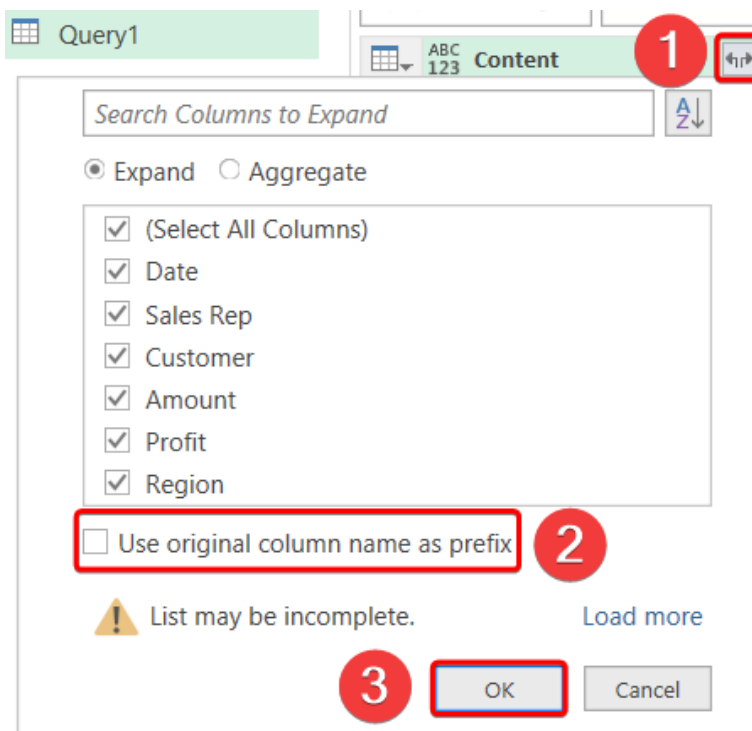
10. Click the Expand Button

- Once clicked, a dialog box will appear showing the fields (columns) you can extract.

11. Uncheck the Prefix Option

- You'll see a checkbox:
"Use original column name as prefix"

12. Uncheck (deselect) this option.



Why?

- If this box remains checked, column names will include the prefix of the original column, like:

Content.Date

Content.SalesRep

Content.Customer

Content.Amount

- This can clutter your column names unnecessarily.

Example Without Unchecking:

- Content.Date, Content.Amount, etc.

Example After Unchecking:

- Date, SalesRep, Customer, Amount

This results in **cleaner and more meaningful column names**.

1. Click OK

- After unchecking the option, click **OK** to expand the selected fields.

	ABC 123 Date	ABC 123 Sales Rep	ABC 123 Customer	ABC 123 Amount	ABC 123 Profit	ABC 123 Region	ABC 123 Name
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	01-Jan-2022 12:00:00 AM	Watson	Namint Enterprises	12850	5380	North	Table_2022
2	09-Jan-2022 12:00:00 AM	D'Souza	VCC	14400	5340	East	Table_2022
3	10-Jan-2022 12:00:00 AM	Roma	Jindle Power Works	14700	4660	South	Table_2022
4	14-Jan-2022 12:00:00 AM	Mark	MNTL	13350	5600	North	Table_2022
5	16-Jan-2022 12:00:00 AM	Steve	VCC	13850	4460	West	Table_2022
6	21-Jan-2022 12:00:00 AM	Vector	Jindle Power Works	11200	3420	South	Table_2022
7	21-Jan-2022 12:00:00 AM	Veronica	Data Tronics	13500	3660	North	Table_2022
8	24-Jan-2022 12:00:00 AM	Watson	Namint Enterprises	13350	3520	East	Table_2022
9	27-Jan-2022 12:00:00 AM	D'Souza	White Associates	14500	3580	East	Table_2022
10	29-Jan-2022 12:00:00 AM	Roma	VCC	14500	4180	East	Table_2022
11	30-Jan-2022 12:00:00 AM	Mark	Namint Enterprises	13150	2400	East	Table_2022
12	01-Feb-2022 12:00:00 AM	Steve	Jindle Power Works	12250	5280	West	Table_2022
13	01-Feb-2022 12:00:00 AM	Vector	White Associates	14150	5120	West	Table_2022
14	03-Feb-2022 12:00:00 AM	Veronica	Shah Associates	13600	3580	North	Table_2022
15	04-Feb-2022 12:00:00 AM	Ramesh	VCC	12300	2560	West	Table_2022
16	06-Feb-2022 12:00:00 AM	Bruce	Namint Enterprises	11950	3320	South	Table_2022
17	06-Feb-2022 12:00:00 AM	Varsha	Good Fly	11050	3740	North	Table_2022
18	14-Feb-2022 12:00:00 AM	Mark	Shyam & Sharma Co	11500	3220	East	Table_2022
19	14-Feb-2022 12:00:00 AM	Rajat	Data Tronics	11450	3800	East	Table_2022
20	17-Feb-2022 12:00:00 AM	Veronica	Shyam & Sharma Co	11050	3120	West	Table_2022
21	17-Feb-2022 12:00:00 AM	James	India Trotters	10800	2660	West	Table_2022
22	21-Feb-2022 12:00:00 AM	Anshika	Sharma & Co	12400	4980	East	Table_2022
23	22-Feb-2022 12:00:00 AM	Abhay	Good Fly	12550	2360	North	Table_2022
24	25-Feb-2022 12:00:00 AM	Varsha	Boston Consultants	13900	3020	East	Table_2022
25	04-Mar-2022 12:00:00 AM	Bruce	Jindle Power Works	12100	2300	West	Table_2022

13. Reordering and Step Review

- After dragging and reordering columns in Power Query, always check the **Applied Steps** pane to ensure no unnecessary steps have been added.
- This pane shows each transformation step that has been applied.
- Reviewing steps helps maintain a clean and efficient query process.

14. Identifying Data Type Issues

- Sometimes, even after importing or cleaning data, the **correct data types are not automatically applied**.
- Common symptoms:
 - **Dates** do not show a calendar icon (meaning they're not recognized as proper date types).
 - **Numbers** are not behaving like numeric fields (e.g., can't sum, no proper formatting).
 - Columns appear as **Text**, even though they contain numbers or dates.

15. Fixing Data Types Efficiently (Without Manual Changes)

The Wrong Way:

- Changing each column's data type manually by clicking one by one — inefficient and time-consuming.

The Right Way:

1. Select All Columns:

- Click anywhere in the table, then press Ctrl + A to select all data.
- The entire table will highlight in green.

2. Detect Data Types Automatically:

- Go to the Transform tab.
- Click Detect Data Type.
- Power Query will now assign the appropriate types:
 - Dates will show a calendar icon.
 - Numbers will display the 123 icons.
 - Text fields will stay as text.

16. Correcting Date Format

- Sometimes the date column is detected as Date/Time, even if you only want the Date.
- **To fix this:**
 - Click the **data type icon** next to the column name (currently showing Date/Time).
 - Change it to **Date** from the dropdown.

The screenshot shows the Power Query Editor interface. The main area displays a table with columns: Name, Date, Sales Rep, Customer, Amount, Profit, and Region. The 'Transform' tab is active, and the 'Data Type' dropdown is open, showing options like Text, Date, Number, etc. A red box highlights the 'Data Type' dropdown, and a red circle highlights the 'Changed Type' step in the 'APPLIED STEPS' pane on the right.

17. Managing 'Change Type' Steps

Problem:

- When you change data types of multiple times, Power Query adds multiple **Changed Type** steps, which can clutter your Applied Steps pane.

Solution:

- When changing a column's type again:
 - Power Query asks: **"Replace current step or add a new one?"**
 - Choose **"Replace current step"** to keep your query clean and efficient.
 - This avoids having several **Changed Type** steps for the same column.

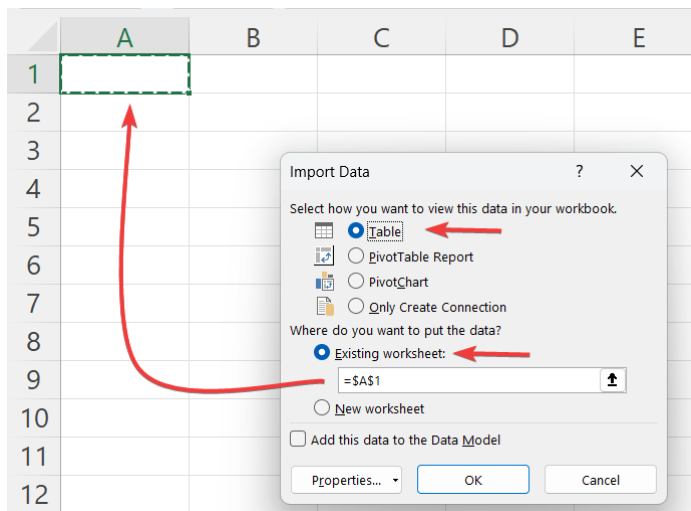
18. Loading Data into Excel

- Once the data is cleaned and the data types are corrected:

Option	Description
Close & Load	Directly loads the data into a new worksheet as a table.
Close & Load To...	Gives you options to load: - Into an existing worksheet Into a table As a PivotTable Only to the Data Model

18. Load the Final Data:

- Click **Close & Load** to load the combined data into Excel as a new table.



19. Load Data Using "Close & Load To"

- After preparing the data in Power Query, go to **Home → Close & Load To**.
- Choose **New Worksheet** and select the **Table** option.
- Click **OK** to load the query result as a table in a new sheet.

20. Understanding the Table Name

- The table is named after the query (e.g., **Query1**).
- You can rename the table and the query to something more meaningful, like **All Data** or **Combined**

Data.

- To rename the query:
 - Double-click on the query name from the Queries panel.
 - Change the name in the query editor.
 - Click **Close & Load** again.

Common Issue: Row Count Keeps Increasing After Refresh

Problem Observation

- Initial row count (e.g., **397 rows**) is correct.

- After clicking **Refresh**, the row count increases significantly (e.g., 795 → 1192 rows).
- The more you refresh, the more rows are added.

Cause of the Issue

- This happens because:
 - The **same workbook** (Excel file) contains both:
 - The **source tables** (e.g., 2022, 2023, 2024).
 - The **output query table** (e.g., All Data).
 - Power Query is set to **detect all tables** in the current workbook and append them.
 - Hence, it also includes the **"All Data" table itself** as a source, leading to duplication every time you refresh.
- **Logical Understanding**
- Power Query is appending all tables in the workbook, including the one it is outputting to — creating a loop.

	A	B	C	D	E	F	G	H	I
1	Name	Date	Sales Rep	Customer	Amount	Profit	Region		
2	Table_2022	01-01-2022 00:00	Watson	Namint Enterprises	12850	5380	North		
3	Table_2022	09-01-2022 00:00	D'Souza	VCC	14400	5340	East		
4	Table_2022	10-01-2022 00:00	Roma	Jindle Power Works	14700	4660	South		
5	Table_2022	14-01-2022 00:00	Mark	MNTL	13350	5600	North		
6	Table_2022	16-01-2022 00:00	Steve	VCC	13850	4460	West		
7	Table_2022	21-01-2022 00:00	Vector	Jindle Power Works	11200	3420	South		
8	Table_2022	21-01-2022 00:00	Veronica	Data Tronics	13500	3660	North		
9	Table_2022	24-01-2022 00:00	Watson	Namint Enterprises	13350	3520	East		
10	Table_2022	27-01-2022 00:00	D'Souza	White Associates	14500	3580	East		
11	Table_2022	29-01-2022 00:00	Roma	VCC	14500	4180	East		
12	Table_2022	30-01-2022 00:00	Mark	Namint Enterprises	13150	2400	East		
13	Table_2022	01-02-2022 00:00	Steve	Jindle Power Works	12250	5280	West		
14	Table_2022	01-02-2022 00:00	Vector	White Associates	14150	5120	West		
15	Table_2022	03-02-2022 00:00	Veronica	Shah Associates	13600	3580	North		
16	Table_2022	04-02-2022 00:00	Ramesh	VCC	12300	2560	West		
17	Table_2022	06-02-2022 00:00	Bruce	Namint Enterprises	11950	3320	South		
18	Table_2022	06-02-2022 00:00	Varsha	Good Fly	11050	3740	North		
19	Table_2022	14-02-2022 00:00	Mark	Shyam & Sharma Co	11500	3220	East		
20	Table_2022	14-02-2022 00:00	Rajat	Data Tronics	11450	3800	East		
21	Table_2022	17-02-2022 00:00	Veronica	Shyam & Sharma Co	11050	3120	West		
22	Table_2022	17-02-2022 00:00	James	India Trotters	10800	2660	West		
23	Table_2022	21-02-2022 00:00	Anshika	Sharma & Co	12400	4980	East		
24	Table_2022	22-02-2022 00:00	Abhay	Good Fly	12550	2360	North		
25	Table_2022	25-02-2022 00:00	Varsha	Boston Consultants	13900	3020	East		
26	Table_2022	04-03-2022 00:00	Bruce	Jindle Power Works	12100	2300	West		
27	Table_2022	05-03-2022 00:00	Ramesh	VCC	11700	5780	East		
28	Table_2022	05-03-2022 00:00	James	Good Fly	14200	2460	North		
29	Table_2022	22-03-2022 00:00	Charley	India Trotters	14800	2820	North		
30	Table_2022	13-03-2022 00:00	Abhay	India Trotters	11800	5040	East		

<	>	2022	2023	2024	2025	Query1	+
---	---	------	------	------	------	--------	---

Queries & Connections

Queries
Connections

1 query

Query1

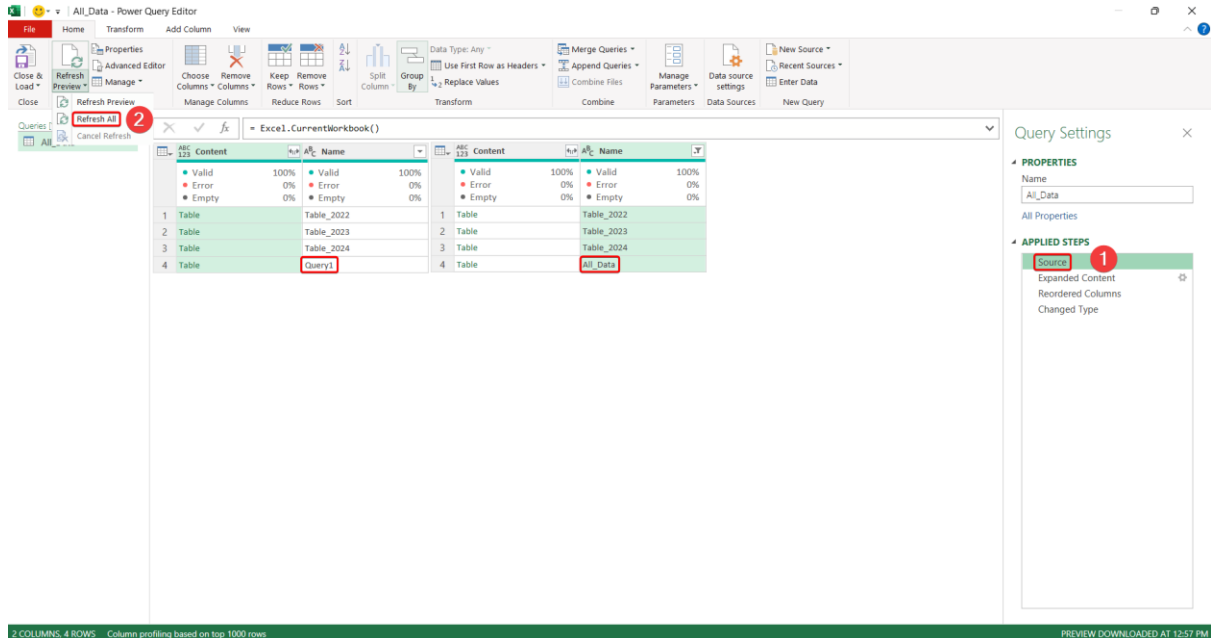
1,192 rows loaded.

Copy
Paste
Edit
Delete
Rename
Refresh
Load To...
Duplicate
Reference
Merge
Append
Export Connection File...
Move To Group
Move Up
Move Down
Show the peek
Properties...

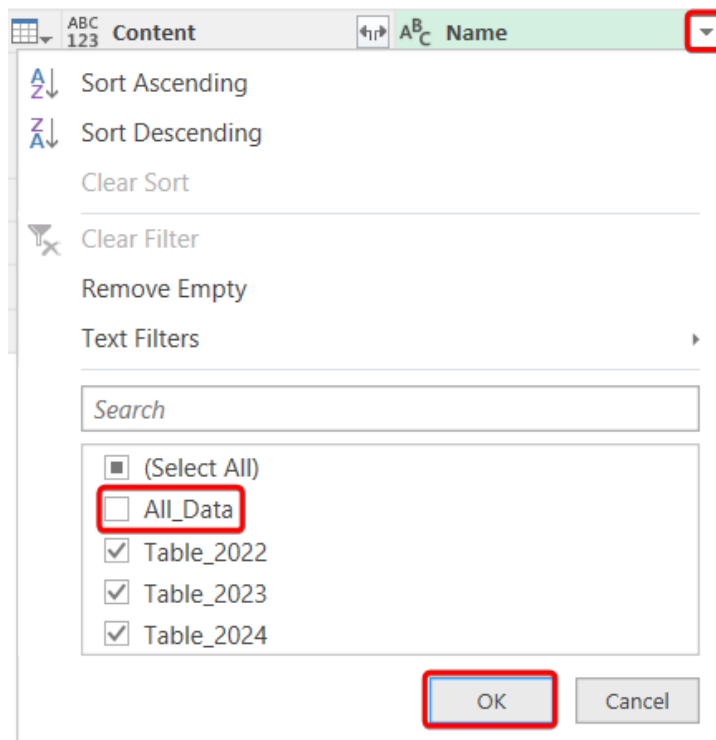
Solution: Exclude Output Table from Being Appended

How to fix:

1. Go to **Power Query Editor**.
2. Select the **Source** step.



1. Click the **Filter** dropdown on the table name column.
2. **Uncheck** the "All Data" table (or whichever is your query output table).
3. Click **OK**.



4. Click the Filter dropdown on the table name column.
5. Uncheck the "All_Data" table (or Whichever is your query output table).
6. Click OK.
7. Click **Close & Load**.

Now, when you refresh:

- The row count remains stable.
- No more unwanted duplications.
- Only actual source tables (e.g., 2022, 2023, 2024) are considered.

Adding New Tables in Future (e.g., 2025)

Instructions:

- Add a new worksheet and paste your data.
- Do not press Ctrl + A (this may include blank spaces or unrelated cells).
- Instead, click anywhere inside your data and press Ctrl + T to convert it to a table.
- Go back to your final query sheet and click Refresh.

The new table (e.g., 2025) will be automatically appended.

Important Tips and Warnings

1. Avoid Blank Rows or Spaces:
 - Never leave an empty row between the data and the added records.
 - Power Query treats it as a separate block and won't include it unless converted into a table again.
 2. Table Formatting Check:
 - When adding new records, ensure they are within the same table.
 - You'll notice the table color band extends to the new row – this confirms it's part of the table.
 3. Adding Data to Existing Table:
 - You can add data directly to the bottom of the existing table (e.g., 2022).
-

- Upon refreshing, those new rows will be included in the combined output automatically.

4. Don't Include Output Table as Input Again:

- This is the most common mistake.
- If you load the query result table inside the same workbook without filtering it out, it will keep appending to itself.

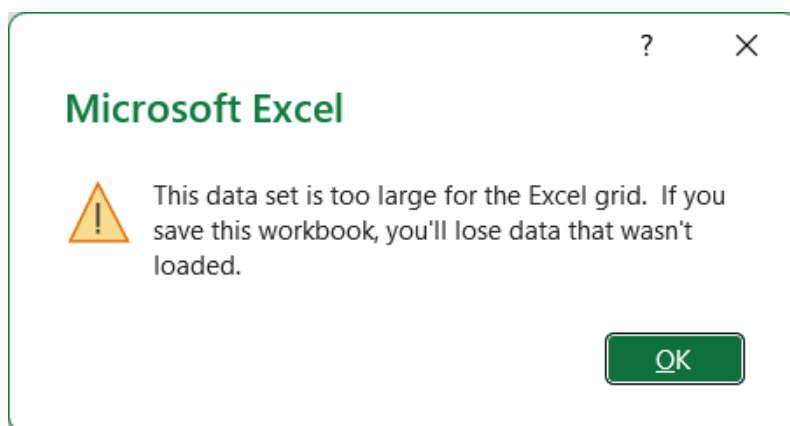
Chapter 3: Handling Large Files in Excel Using Power Query

Introduction

When dealing with **large datasets** (such as a CSV file with more than 1 million rows), Excel's default capabilities become a limitation. In this chapter, you'll learn **how to overcome Excel's row limit** using **Power Query** and how to analyse large data efficiently using **Pivot Tables**, without needing external tools like Power BI.

Problem: Excel Has a Row Limit

- Microsoft Excel can handle **only up to 1,048,576 rows** per worksheet.
- If your file (e.g., a CSV) contains **1.5 million rows**, Excel will:
 - Load only the **first 1,048,576 rows**
 - **Truncate** the remaining rows without warning
 - Potentially lose important data if the file is saved after opening



Important: Never save such a file immediately after opening it in Excel — doing so will result in permanent data loss for the truncated rows.

Example Scenario

- You receive a CSV file with **1.5 million rows** (such as accident data).
- You open it in Excel, and it looks fine — but only the first 1,048,576 rows are visible.
- The rest of the data is missing.
- You need to generate a **summary report** (e.g., using Pivot Tables), but Excel won't let you load all the rows.

Solution: Use Power Query

Power Query is capable of handling datasets **larger than Excel's visible grid**. It allows you to **transform, clean, and load** data into Excel's **Data Model** instead of the worksheet.

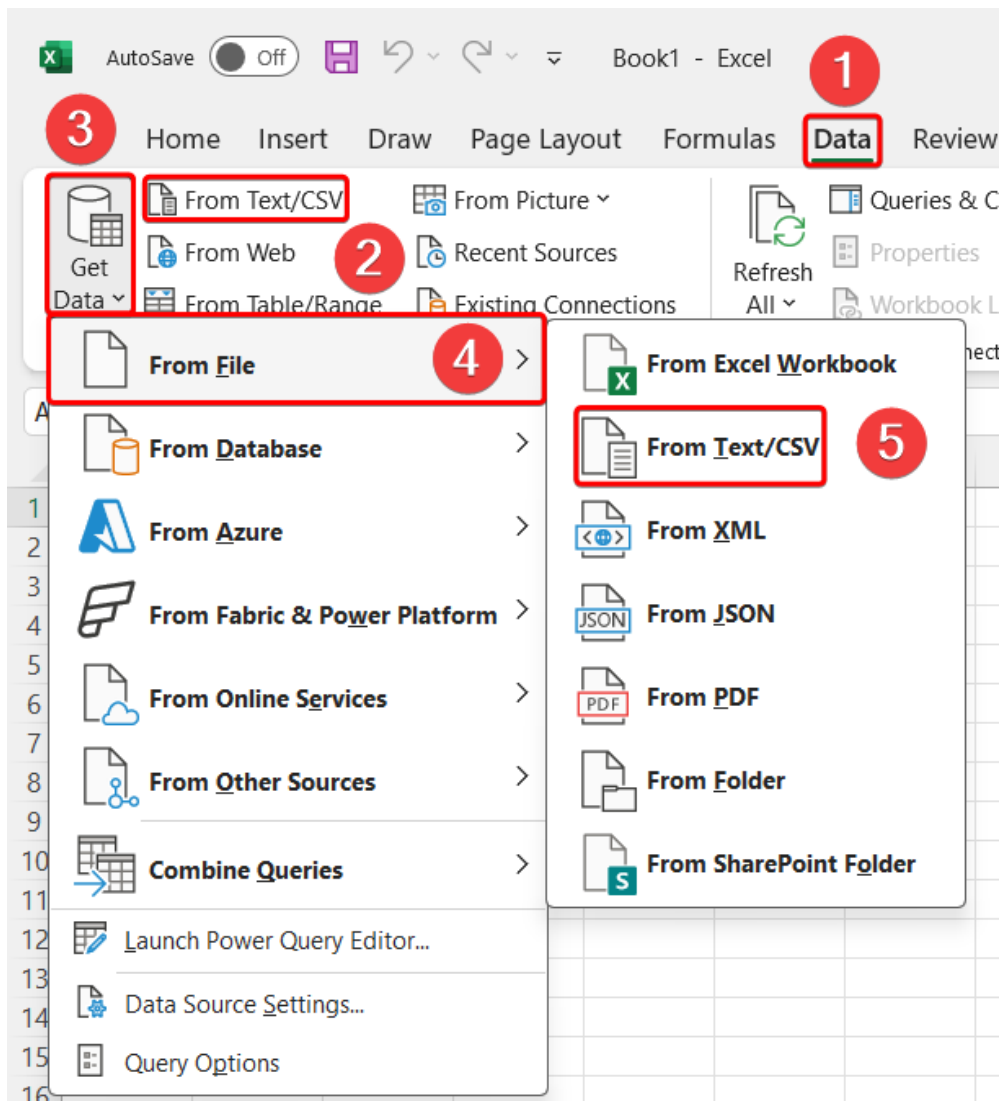
This approach helps:

- Avoid Excel's row limit
- Work efficiently with large files
- Generate powerful reports using Pivot Tables

Step-by-Step Guide: Handling Large Files

❖ Step 1: Import the Large File

1. Go to the **Data** tab.
2. Select **From Text/CSV** (or navigate via **Get Data > From File > From Text/CSV**).



3. Select your large CSV file.
4. Click **Import**.

Large Data Set Practice File.csv

File Origin: 65001: Unicode (UTF-8) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Accident_Index	Location_Easting_OSGR	Location_Northing_OSGR	Longitude	Latitude	Police_Force	Accident_Severity	Number
200501BS00001	525680	178240	-0.19117	51.489096	1	2	
200501BS00002	524170	181650	-0.211708	51.520075	1	3	
200501BS00003	524520	182240	-0.206458	51.525301	1	3	
200501BS00004	526900	177530	-0.173862	51.482442	1	3	
200501BS00005	528060	179040	-0.156618	51.495752	1	3	
200501BS00006	524770	181160	-0.203238	51.51554	1	3	
200501BS00007	524220	180830	-0.211277	51.512695	1	3	
200501BS00009	525890	179710	-0.187623	51.50226	1	3	
200501BS00010	527350	177650	-0.167342	51.48342	1	3	
200501BS00011	524550	180810	-0.206531	51.512443	1	3	
200501BS00012	526240	178900	-0.182872	51.494902	1	3	
200501BS00014	526170	177690	-0.184312	51.484044	1	3	
200501BS00015	525590	178520	-0.192366	51.491632	1	3	
200501BS00016	527990	178690	-0.157753	51.492622	1	3	
200501BS00017	526700	178970	-0.176224	51.495429	1	3	
200501BS00018	526460	177460	-0.18022	51.481912	1	3	
200501BS00019	524680	179450	-0.205139	51.500191	1	2	
200501BS00020	527000	179020	-0.171887	51.495811	1	3	
200501BS00021	527810	178010	-0.16059	51.486552	1	3	
200501BS00022	526790	178980	-0.174925	51.495498	1	2	

Click on Transform Data

Load Transform Data Cancel

Excel will use Power Query to show a **preview of the first 200 rows** for data validation.

❖ **Step 2: Do Not Click 'Load'**

- Avoid using "Close & Load" — this will try to load the full dataset into the Excel grid, which will fail.
- Instead, choose "Transform Data" to enter Power Query Editor.

❖ **Step 3: Transform the Data (Optional)**

Once in Power Query:

- Review the columns and headers.
- Use transformation tools to:
 - Remove unwanted columns
 - Detect and change data types (e.g., convert text to date or number)
 - Rename columns if needed
- Power Query automatically applies steps like:
 - Promote Headers
 - Change Data Type

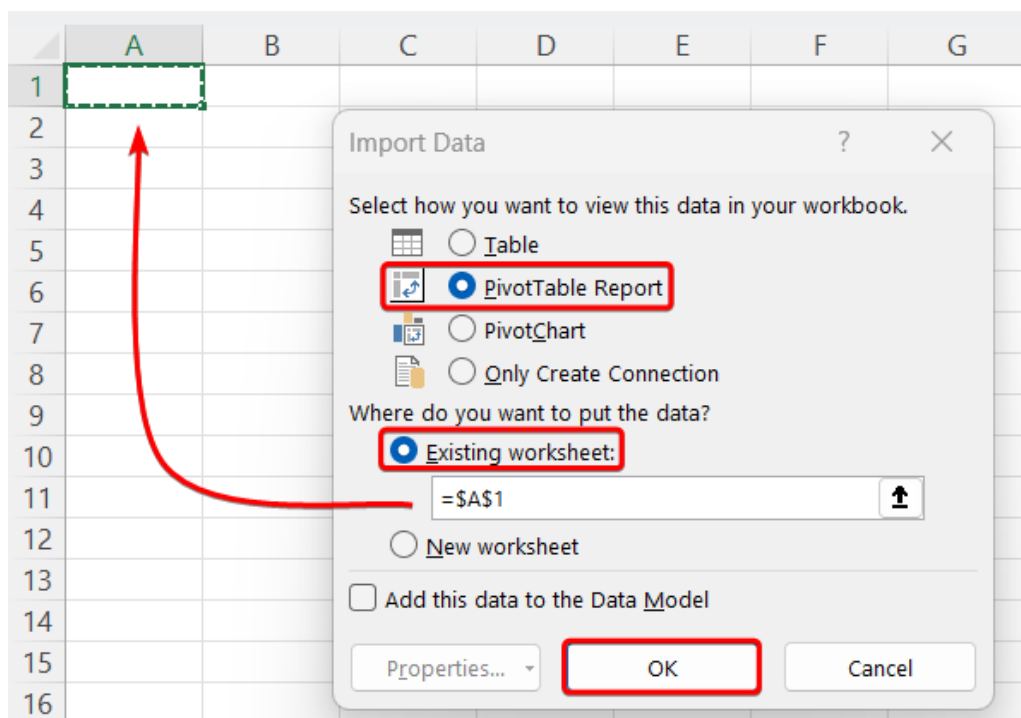
You can view these steps in the **Applied Steps** pane.

❖ **Step 4: Load Data to a Pivot Table**

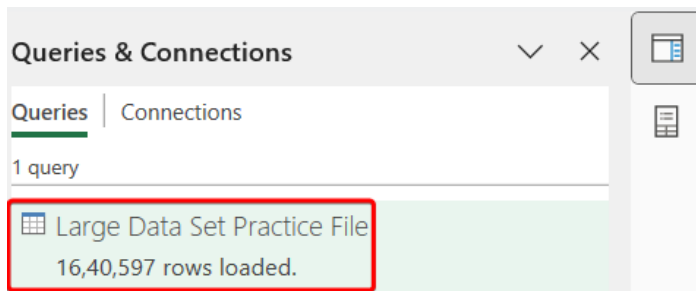
Since we cannot load millions of rows into Excel directly, we use the **Data Model**.

Here's how:

1. Click **Close & Load To...**
2. Select:
 - Only Create Connection
 - Add this data to the Data Model
3. Choose **Pivot Table Report**
4. Choose where to place the Pivot Table:
 - **New Worksheet**
 - Or an **Existing Worksheet** (recommended if it's empty)



This method loads the data in the background (to the data model), not into Excel's grid, bypassing row limits.



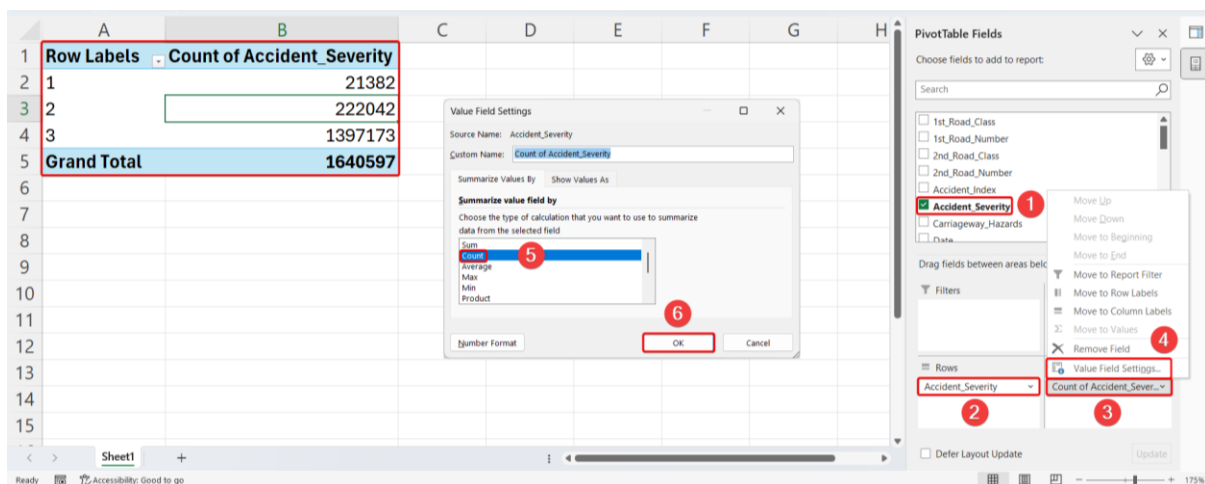
Step 5: Creating a Summary Report (Using Pivot Table)

Now that the data is available in the Data Model, build a **Pivot Table** to summarize your data.

Example: Accident Severity Report

You can create a report showing **accident severity levels**:

- Drag **Accident Severity** into the **Rows** section:
 - 1 = Low/Natural accidents
 - 2 = Medium severity
 - 3 = High/Critical severity
- Drag **Accident ID** into the **Values** section:
 - Set the calculation to **Count**, not **Sum** (use Value Field Settings)



1. Drag and drop Accident_Severity into the Rows Section.
2. Then, Drag and drop Accident_serverity Again into the Values section.
3. Now, click the down arrow in the Value field. and then select Value Field Settings from the menu.
4. Set the calculation to Count, not Sum (use Value Field Settings) and click Ok.

Chapter 4: Appending Data from Multiple Excel Workbooks

❖ Introduction

- This is one of the most **common challenges faced** by Excel users.
- The problem: Users often receive **data in multiple Excel files** (workbooks)—for example:
 - Monthly or yearly bank statements.
 - Financial reports.
 - Any dataset split across multiple years/files.
- Manual copying and pasting is **inefficient** and **error prone**.
- Power Query can automate this using the **“Folder” option** to import and append all Excel files from a given folder.

❖ Real-World Scenario

- In many companies, someone is manually appending data from hundreds of files daily.
- A Power Query expert mentioned in a podcast that:

A company hired a full-time assistant just to append Excel data manually. This could have been done using Power Query in minutes.

- This chapter teaches how to automate that process completely.

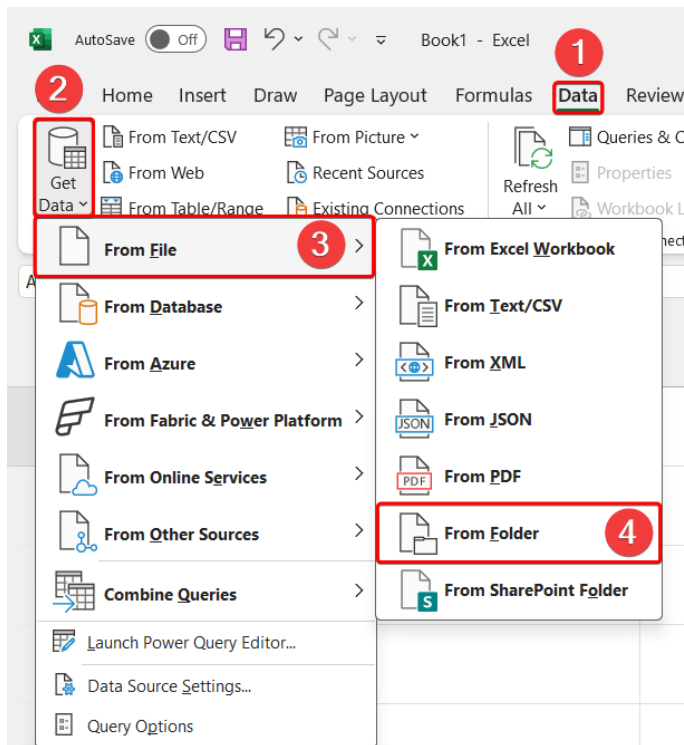
Steps to Append Data from Multiple Excel Workbooks

1 Setup a Clean Environment

- **Important:** Do **not create the master file inside the same folder** where your source files are.
- Open a **new blank Excel workbook** from the desktop or outside the data folder.
- Do **not** right-click → New → Excel in the data folder (can cause loop issues).

2 Connect to Folder

1. Go to the **Data tab → Get Data → From File → From Folder**.



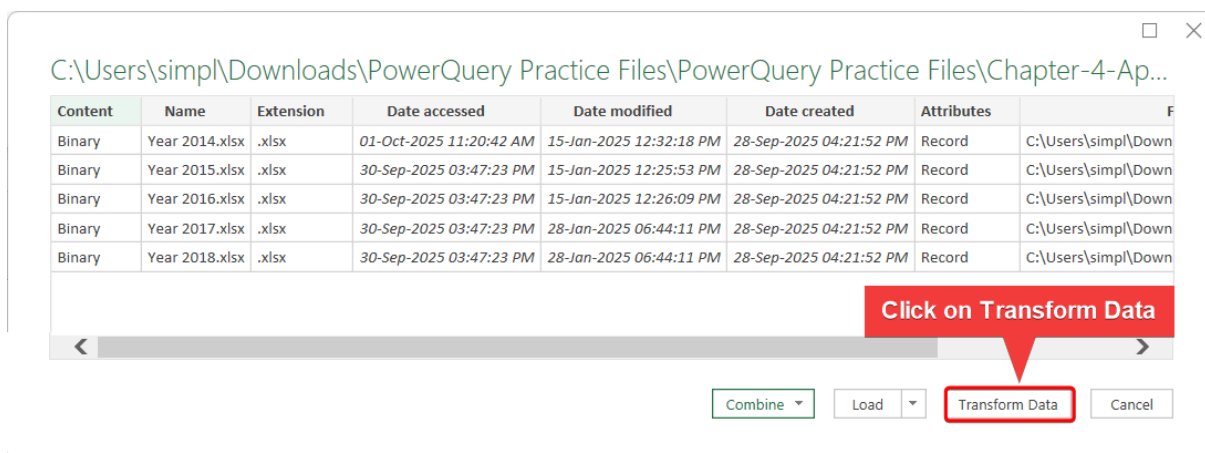
2. Select the **folder** containing all the Excel files (e.g., files for 2014, 2015, 2016, etc.).

3. A preview appears showing:

- File name
- Date modified
- File path
- A “**Content**” column (with binary file contents)

3 Load to Power Query (Transform Data)

- Click “**Transform Data**”, not “Load”.



- In the Power Query Editor:

1. Select only the columns: **Content** and **Name**.

- Use Ctrl + Click to select both.
- Right-click → **Remove Other Columns**

	Content		Name	
	- Valid 100% - Error 0% - Empty 0%		- Valid 100% - Error 0% - Empty 0%	Copy Remove Columns Remove Other Columns Add Column From Examples... Remove Duplicates Remove Errors Fill Change Type
1	Binary		Year 2014.xlsx	
2	Binary		Year 2015.xlsx	
3	Binary		Year 2016.xlsx	
4	Binary		Year 2017.xlsx	
5	Binary		Year 2018.xlsx	

2. Rearrange columns if needed (e.g., Name first, Content second).

Extract Excel Workbook Content

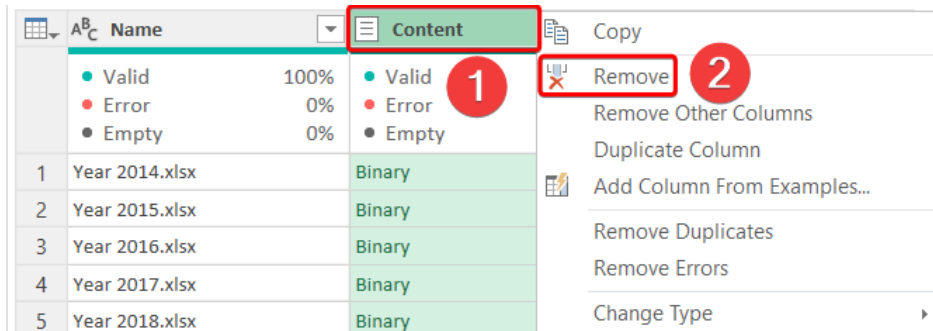
- Add a custom column using **Add Column** → **Custom Column**.
- Use this formula to extract content from the binary files:

Excel.Workbook([Content])

The screenshot shows the Power Query Editor interface. The 'Add Column' tab is selected in the ribbon. The 'Custom Column' dialog box is open, showing the formula 'Excel.Workbook([Content])' and the new column name 'Custom'. The 'Available columns' list on the right shows 'Name' and 'Content'. The 'Query Settings' pane on the right shows the 'Applied Steps' list with 'Added Custom' at the bottom. The main data table shows columns 'Name', 'Content', and 'Custom'.

1. First, select the Content column.
2. Next, click on the add Column tab.

3. Then, within the tab, click on the Add Column option.
4. You can enter any name for the column in the New Column Name Field.
5. Use this formula to extract content from the binary files: Excel.Workbook([Content])



- Select the Content column, then right click with your mouse and click on the Remove option. The Content column will then be removed.

5 Understand the Hierarchy

- Excel files have this structure:

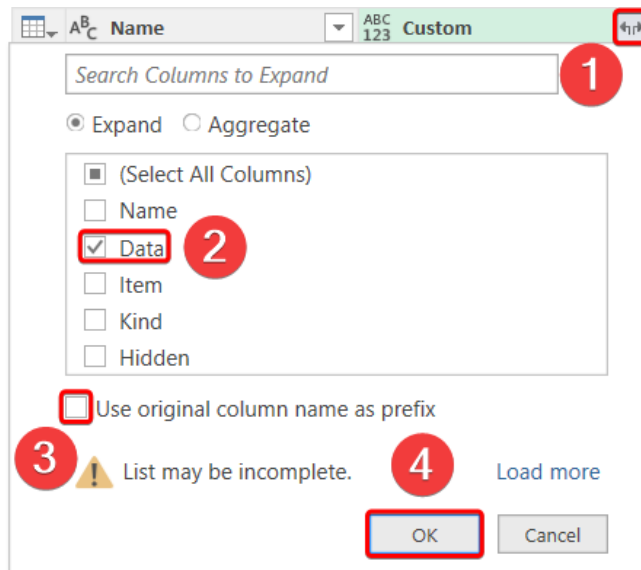
Workbook → Sheets → Tables

So, the data needs to be **navigated step-by-step**:

- First extract the workbook
- Then extract the sheet
- Then extract the data table

6 Expand the Extracted Data

1. After applying the custom column, preview the new column (click next to the word “Table”).
2. You’ll see:
 - Sheet name
 - Item
 - Kind (Sheet/Table)
 - Hidden (Yes/No)
3. Keep only the column “**Data**” (which holds the actual table).
 - Uncheck others (Item, Kind, Hidden, etc.)



7 Promote Headers in All Tables

Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9	Column10	Column11
CA-2014-140795	02-Jan-2014	First Class	BD-11500	Bradley Drucker	Consumer	United States	Green Bay	Wisconsin	54302	Wisconsin
CA-2014-104269	03-Jan-2014	Second Class	DB-13060	Dave Brooks	Consumer	United States	Seattle	Washington	98115	Washington
US-2014-143707	03-Jan-2014	Standard Class	HR-14770	Hallie Redmond	Home Office	United States	New York City	New York	10035	East
CA-2014-113880	03-Jan-2014	Standard Class	VF-21715	Vicky Freymann	Home Office	United States	Elmhurst	Illinois	60126	Central
CA-2014-113880	03-Jan-2014	Standard Class	VF-21715	Vicky Freymann	Home Office	United States	Elmhurst	Illinois	60126	Central
CA-2014-111009	03-Jan-2014	Standard Class	SC-20380	Shahid Collister	Consumer	United States	El Paso	Texas	79907	Central
CA-2014-111009	03-Jan-2014	Standard Class	SC-20380	Shahid Collister	Consumer	United States	El Paso	Texas	79907	Central
CA-2014-111009	03-Jan-2014	Standard Class	SC-20380	Shahid Collister	Consumer	United States	El Paso	Texas	79907	Central
CA-2014-111009	03-Jan-2014	Standard Class	SC-20380	Shahid Collister	Consumer	United States	El Paso	Texas	79907	Central
CA-2014-168312	03-Jan-2014	Standard Class	GW-14605	Gulietta Weimer	Consumer	United States	Houston	Texas	77036	Central
CA-2014-168312	03-Jan-2014	Standard Class	GW-14605	Gulietta Weimer	Consumer	United States	Houston	Texas	77036	Central
US-2014-157021	04-Jan-2014	Second Class	KM-16720	Kunst Miller	Consumer	United States	Vallejo	California	94591	West
US-2014-157021	04-Jan-2014	Second Class	KM-16720	Kunst Miller	Consumer	United States	Vallejo	California	94591	West
CA-2014-138359	04-Jan-2014	Standard Class	KH-16330	Katharine Harms	Corporate	United States	Revere	Massachusetts	2151	East

1. Click on the blank space, and then you'll see the table displayed below.

2. Each table, the headers are shown as Column1, Column2, etc., which is incorrect.

Problem: Headers from individual files are coming as the **first row of data**.

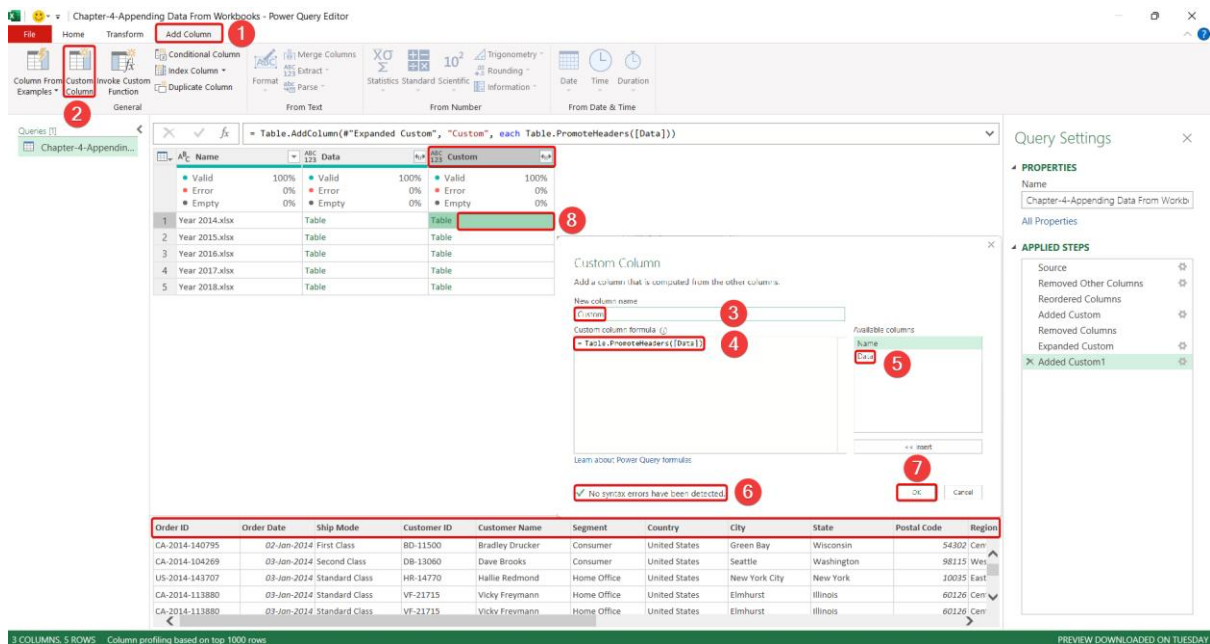
- To fix:
 - Add another **Custom Column** with this formula:

Table.PromoteHeaders([Data])

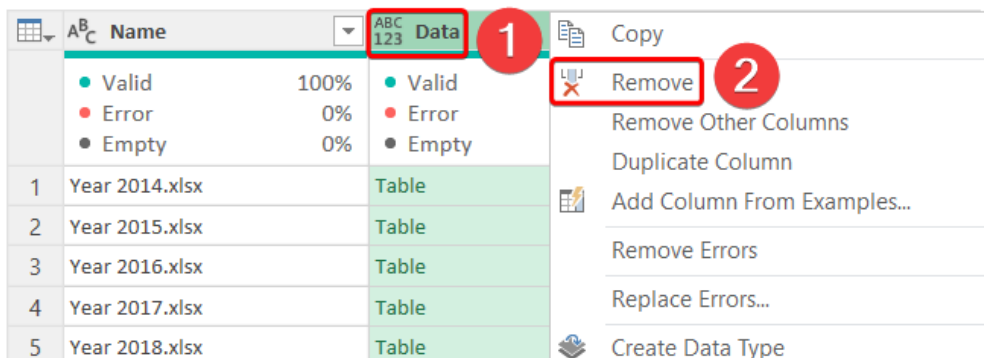
- This will apply headers uniformly across all tables.

4. Click on the 'Add Column' Tab

- Go to the top menu and click on the "Add Column" tab.
- Click on 'Custom Column'
- In the **General** group, click on "Custom Column".
- Name the Column**
 - In the dialog box, enter a name for your new column (e.g., PromotedHeaders).
- Enter the Formula**
 - In the formula box, type the following code:
- Click OK**
 - After entering the formula, click **OK** to create the custom column.
- Once you click OK, the custom column will be added. By Clicking on the blank area be of each table, you can verify that the headers have been correctly applied.



- Select the Data column, then right click and choose Remove.



8 Expand Final Table

- Click the **expand icon** next to the new column.

- Uncheck “Use original column name as prefix”.
- Select all columns to include them in the final table.

Formula Bar: = Table.ExpandTableColumn(#\"Removed Columns1\", \"Custom\", {\"Order ID\", \"Order Date\", \"Ship Mode\", \"Customer ID\", \"Customer Name\", \"Segment\"}, ...)

	Name	Order ID	Order Date	Ship Mode	Customer ID	Customer Name	Segment
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	Year 2014.xlsx	CA-2014-140795	02-Jan-2014	First Class	BD-11500	Bradley Drucker	Consumer
2	Year 2014.xlsx	CA-2014-104269	03-Jan-2014	Second Class	DB-13060	Dave Brooks	Consumer
3	Year 2014.xlsx	US-2014-143707	03-Jan-2014	Standard Class	HR-14770	Hallie Redmond	Home Office
4	Year 2014.xlsx	CA-2014-113880	03-Jan-2014	Standard Class	VF-21715	Vicky Freymann	Home Office
5	Year 2014.xlsx	CA-2014-113880	03-Jan-2014	Standard Class	VF-21715	Vicky Freymann	Home Office

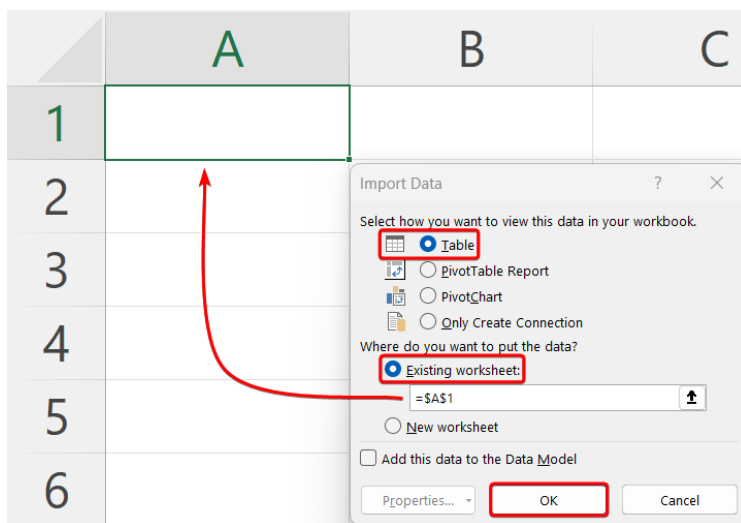
9 Ensure Correct Data Types

1. First, press Ctrl+A to select the entire table.
2. Next, click on the Transform Tab.
3. Within the Transform tab, click on the Detect Data Type option.
4. You can now see that the correct data types have been applied to all the column headers.

- This ensures:
 - Dates become **Date type**
 - Numbers become **Whole/Decimal**
 - Text stays as **Text**

10 Load Data to Excel

- Go to **Home → Close & Load To**
- Load the final table to:



- A new worksheet
- Or an **existing blank space**

Adding More Files Later

- Just copy any new Excel file (e.g., 2018 data) into the **same folder**.
- **Do not** open or edit those files.
- Simply **right-click → Refresh** in Excel.

Name	Order ID	Order Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code
Year 2014.xlsx	CA-2014-140795	02-01-2014	First Class	BD-11500	Bradley Drucker	Consumer	United States	Green Bay	Wisconsin	54302
Year 2014.xlsx	CA-2014-104269	03-01-2014	Second Class	DB-13060	Dave Brooks	Consumer	United States	Seattle	Washington	98115
Year 2014.xlsx	US-2014-143707	03-01-2014	Standard Class	HR-14770	Hallie Redmond	Home Office	United States	New York City	New York	10035
Year 2014.xlsx	CA-2014-113880	03-01-2014	Standard Class	VF-21715	Vicky Freymann	Home Office	United States	Elmhurst	Illinois	60126
Year 2014.xlsx	CA-2014-113880	03-01-2014	Standard Class	VF-21715	Vicky Freymann	Home Office	United States	Elmhurst	Illinois	60126
Year 2014.xlsx	CA-2014-131009	03-01-2014	Standard Class	SC-20380	Shahid Collister	Consumer	United States	El Paso	Texas	79907
Year 2014.xlsx	CA-2014-131009	03-01-2014	Standard Class	SC-20380	Shahid Collister	Consumer	United States	El Paso	Texas	79907
Year 2014.xlsx	CA-2014-131009	03-01-2014	Standard Class	SC-20380	Shahid Collister	Consumer	United States	El Paso	Texas	79907
Year 2014.xlsx	CA-2014-168312	03-01-2014	Standard Class	GW-14605	Giulietta Weimer	Consumer	United States	Houston	Texas	77036
Year 2014.xlsx	CA-2014-168312	03-01-2014	Standard Class	GW-14605	Giulietta Weimer	Consumer	United States	Houston	Texas	77036
Year 2014.xlsx	US-2014-157021	04-01-2014	Second Class	KH-16720	Kunst Miller	Consumer	United States	Vallejo	California	94591
Year 2014.xlsx	CA-2014-138359	04-01-2014	Standard Class	KH-16330	Katharine Harms	Corporate	United States	Revere	Massachusetts	2151
Year 2014.xlsx	CA-2014-138359	04-01-2014	Standard Class	KH-16330	Katharine Harms	Corporate	United States	Revere	Massachusetts	2151
Year 2014.xlsx	CA-2014-131926	06-01-2014	Second Class	DW-13480	Dianna Wilson	Home Office	United States	Lakeville	Minnesota	55044
Year 2014.xlsx	CA-2014-131926	06-01-2014	Second Class	DW-13480	Dianna Wilson	Home Office	United States	Lakeville	Minnesota	55044
Year 2014.xlsx	CA-2014-131926	06-01-2014	Second Class	DW-13480	Dianna Wilson	Home Office	United States	Lakeville	Minnesota	55044
Year 2014.xlsx	CA-2014-131926	06-01-2014	Second Class	DW-13480	Dianna Wilson	Home Office	United States	Lakeville	Minnesota	55044
Year 2014.xlsx	CA-2014-111003	06-01-2014	Standard Class	CR-12625	Corey Roper	Home Office	United States	Lakewood	New Jersey	8701
Year 2014.xlsx	CA-2014-111003	06-01-2014	Standard Class	CR-12625	Corey Roper	Home Office	United States	Lakewood	New Jersey	8701
Year 2014.xlsx	US-2014-165659	06-01-2014	Standard Class	LT-17110	Liz Thompson	Consumer	United States	Little Rock	Arkansas	72209
Year 2014.xlsx	US-2014-165659	06-01-2014	Standard Class	LT-17110	Liz Thompson	Consumer	United States	Little Rock	Arkansas	72209
Year 2014.xlsx	US-2014-157070	06-01-2014	Standard Class	QJ-19255	Quincy Jones	Corporate	United States	Detroit	Michigan	48234
Year 2014.xlsx	US-2014-157070	06-01-2014	Standard Class	QJ-19255	Quincy Jones	Corporate	United States	Detroit	Michigan	48234

- Power Query automatically:
 - Detects new files
 - Extracts and appends their data
 - Updates the master table

Real Case Example in Training

- Many trainees share that:
 - Sometimes new files have **extra columns** (e.g., a new column in 2018).
 - Power Query handles this gracefully:
 - It adds the column in 2018 data.
 - Other rows (e.g., 2014-2017) will just have blanks for that column.
- No issue as long as:
 - Column headers are mostly consistent
 - Date format is uniform

Chapter 5: Appending CSV Files in Excel (Power Query)

Introduction

- Now, we will learn how to append CSV files.
- Why CSV?
 - Many times, data is stored in CSV format (very large datasets – more than 1 million rows).
 - Multiple CSV files can exist in one folder (e.g., sales data from different regions).
- Goal: Append (combine) all CSV files into one table in Power Query.

Important Rules Before Appending

1. All CSV files should have similar structure (same headers).
2. If an extra column is present in one file, Power Query will still append it (new column will be created).
3. Always open a new blank Excel file in a separate folder (don't keep it in the same folder as CSV files).

Example Scenario

- We have 3 CSV files:
 - Central.csv
 - South.csv
 - West.csv

👉 Task: Append them into one table in Excel using Power Query.

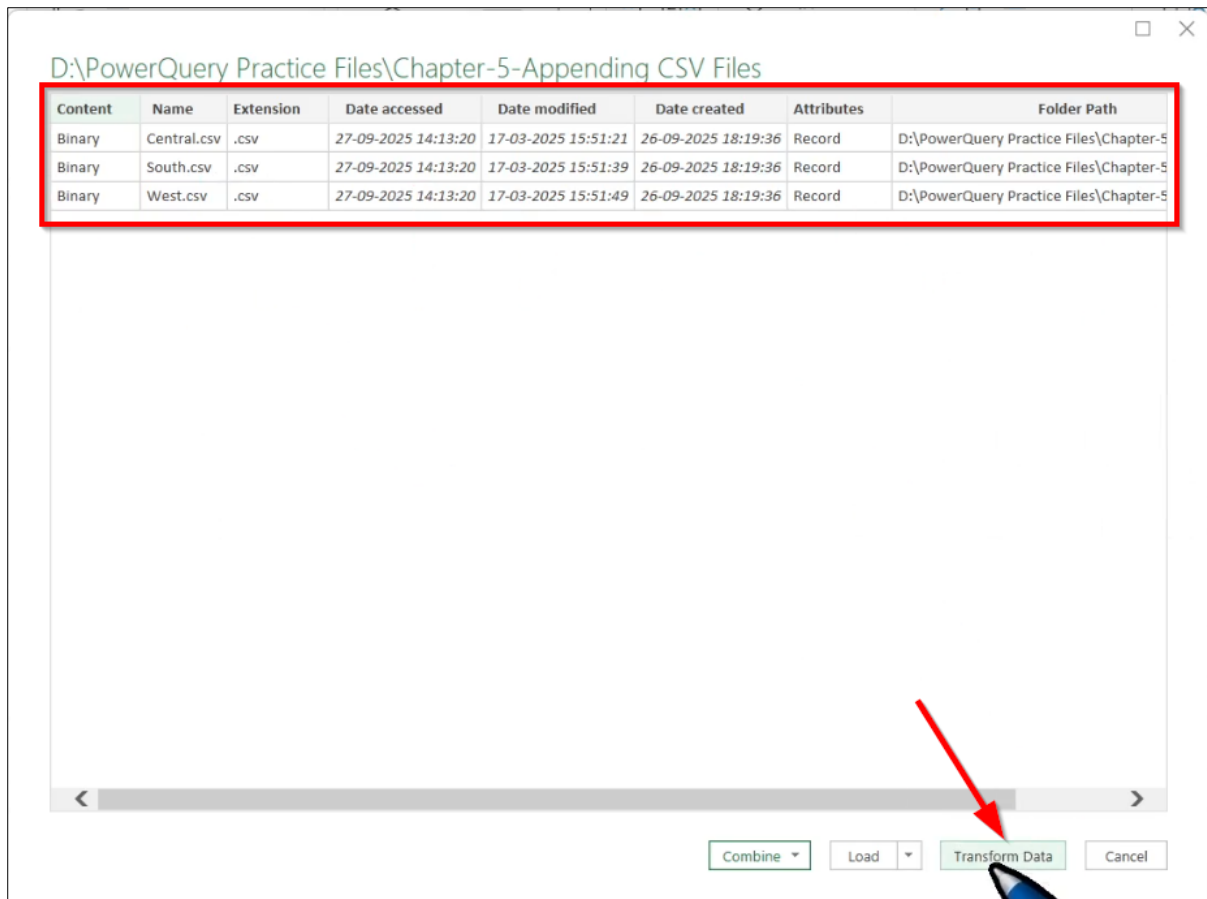
Step-by-Step Process

Step 1: Open Power Query from a Blank File

1. Open a new blank Excel workbook.
2. Go to Data Tab → Get Data → From File → From Folder.
3. Browse and select the folder that contains all your CSV files.

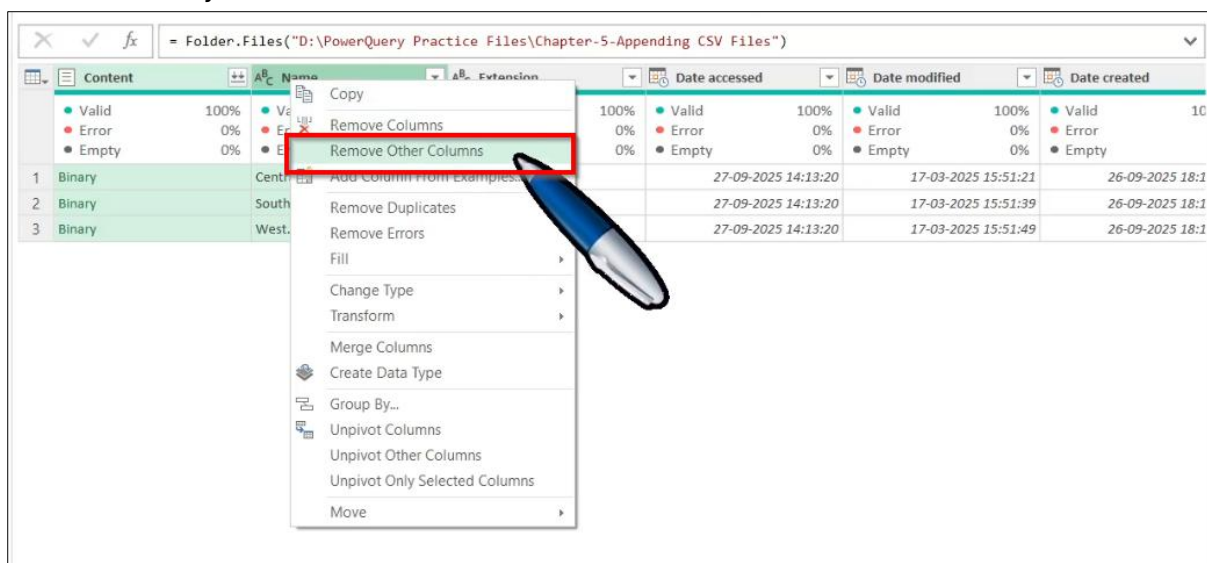
Step 2: Preview the Folder Content

- After selecting the folder, Excel will show a preview:
 - Name (file name)
 - Content (binary file data)
 - Other file properties (extension, date modified, etc.)
- Click **Transform Data** to go into Power Query.



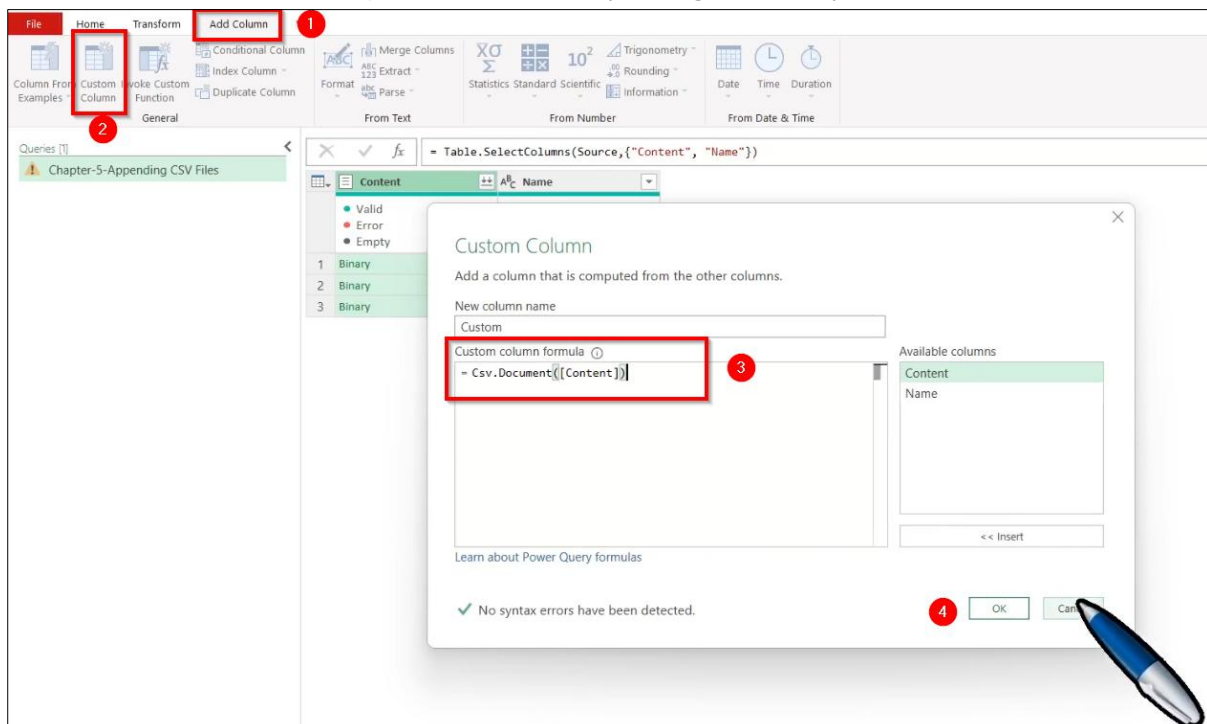
Step 3: Keep Only Required Columns

- In Power Query, you will see columns like:
 - Content** → holds the actual CSV data
 - Name** → file name
 - Other unnecessary columns may exist
- Select only Content and Name columns** (Ctrl + Click to select).
- Right-click → **Remove Other Columns**.
- Now only the two needed columns remain.



Step 4: Extract Data from Each CSV File

- We cannot expand Content directly because it's in binary format.
- Go to Add Column → Custom Column.
- Formula to use: **Csv.Document([Content])**
- This will convert binary CSV file into a table.
- Check the formula carefully (capitalization matters):
 - C → capital, s → small, v → small
 - If you see errors, check if you accidentally wrote Cvs.Document or something incorrect.
- Remove the old binary **Content** column (no longer needed).



Step 5: Promote Headers for Each Table

- Now, each file is converted into a table, but headers are not proper (showing Column1, Column2, etc.).
- Add another Custom Column with formula: **Table.PromoteHeaders([Custom])**
- This step makes the first row as header for each table.

Custom Column

Add a column that is computed from the other columns.

New column name
Custom.1

Custom column formula ⓘ
= Table.PromoteHeaders([Custom])

Available columns
Name
Custom

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

- This will add New Column name **Custom1** with proper headers
- Now remove the old Custom column which now not required.

Step 6: Expand All Tables Together

- Now expand the tables:
 - Click on Expand button (two-arrow icon).
 - Uncheck "Use Original Column Name as Prefix".
 - Click OK.
- All CSV file data is now combined into one table.

ABC Name ABC 123 Custom.1

Search Columns to Expand

Expand Aggregate

(Select All Columns)
ID
Product
Quantity

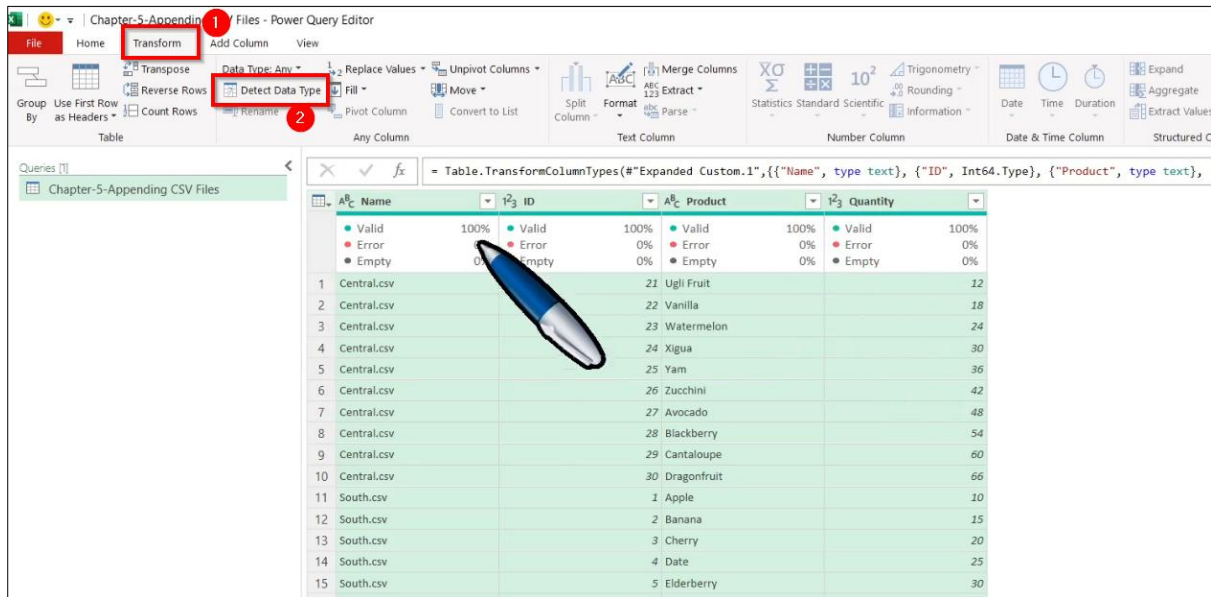
Use original column name as prefix

List may be incomplete. Load more

OK Cancel

Step 7: Fix Data Types

- Select the entire table (Ctrl + A).
- Go to Transform → Detect Data Type.
- Power Query will automatically set each column (Text, Number, Date, etc.).



Step 8: Load Data into Excel

- Once data is clean and combined, click Close & Load.
- Data is loaded into Excel as a table.

Refresh Feature (Dynamic Updates)

- If you add a new CSV file (e.g., East.csv) into the same folder:
 - Simply right-click on the query table in Excel → Refresh.
 - Data from the new file will also be included automatically.

👉 Example: Initially, 30 rows → After adding East.csv (10 rows) → 40 rows after refresh.

Chapter 6: Advanced Grouping in Power Query – Employee Leave Analysis

Introduction

In this chapter, we will learn **advanced grouping in Power Query** to solve a practical problem:

What is Advanced Grouping?

- **Advanced Grouping** allows us to do more than just count or sum.
- We can group data by a column and also keep related details together.
- Example: Group leave data by Date → see **how many employees are on leave** and also **which employees are on leave**.

Problem Statement:

We have a table with employees' names, start dates, and end dates of leaves.

The goal is:

1. Identify how many employees are on leave on a particular date.
2. List the names of employees on leave for each date.

This cannot be solved easily with standard Excel formulas because overlapping leave dates make counting and listing challenging. Power Query simplifies this process.

Data Example

	A	B	C
1	Emp_name	Start Date	End Date
2	Ajit Sharma	01-01-2025	04-01-2025
3	Vijay Singh	01-01-2025	06-01-2025
4	Rani Maurya	13-01-2025	13-01-2025
5	Ajay Kale	12-01-2025	15-01-2025
6	Vishal Sawant	03-01-2025	05-01-2025
7	Sejal Singh	20-01-2025	20-01-2025
8	Arti Pandey	19-01-2025	20-01-2025
9	Vinay Jadhav	23-01-2025	23-01-2025

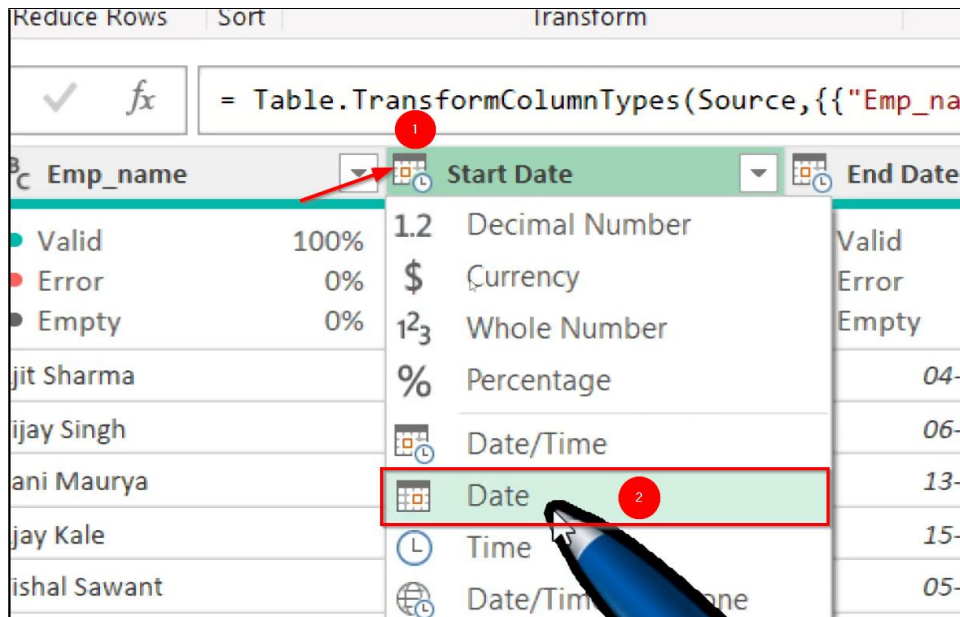
Observation:

- On 3rd Jan, three employees are on leave: Ajit, Vijay, and Vishal.
- Counting manually is time-consuming and error-prone.

Step-by-Step Solution:

Step 1: Load Data into Power Query

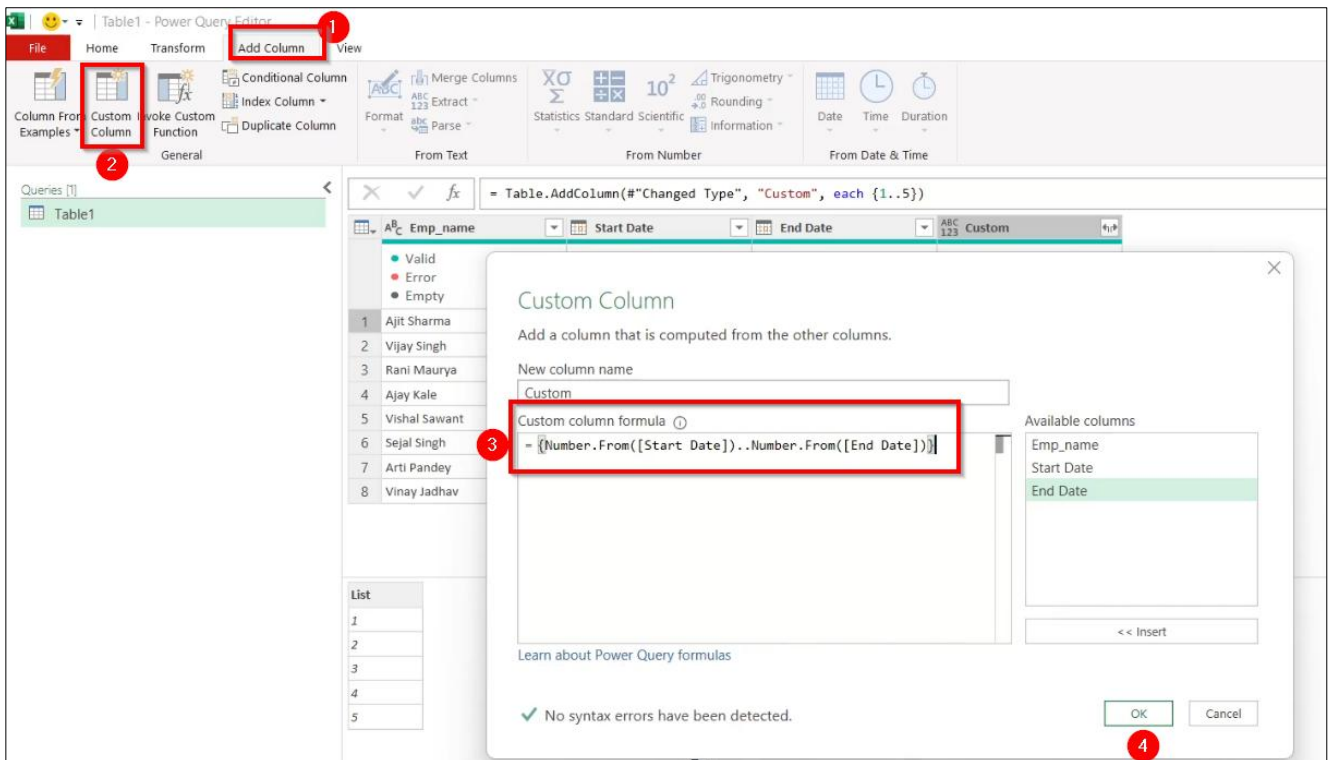
1. Select your table → **Data Tab** → **From Table/Range**.
2. Ensure all date columns are correctly formatted as **Date**:
 - Transform → Data Type → Date
 - If necessary, use **Change Type using Locale**.



Step 2: Generate Date List for Each Employee

We need to expand each employee's leave into all individual dates:

1. **Add Custom Column** → Formula:
 - `{ Number.From([Start Date]) .. Number.From([End Date]) }`
 - Generates a list of all dates between start and end.
 - Example: `{01-01-2025..04-01-2025}` → 01, 02, 03, 04 Jan.



2. Breakdown of the Formula: { Number.From([Start Date]) .. Number.From([End Date]) }

- **[Start Date] and [End Date]**

These are assumed to be date-type columns in your table.

- **Number.From(...)**

This converts a date value into its **serial number** — the number of days since **December 30, 1899** (which is day 0 in Power Query's date system).

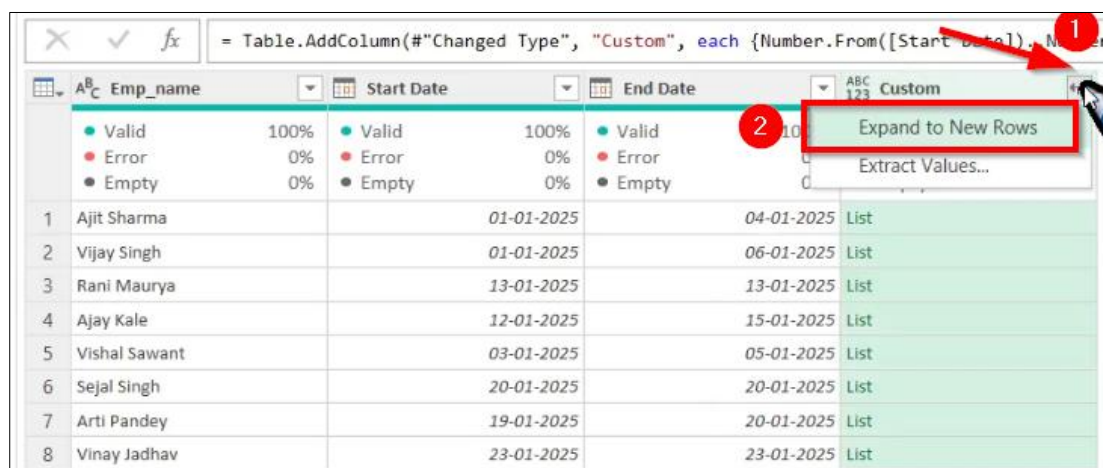
For example:

- `Number.From(#date(2025, 10, 1))` → 46083
- `Number.From(#date(2025, 10, 3))` → 46085

- **{ a .. b }**

This is a **range generator**. It creates a list of numbers from a to b, inclusive.

3. Expand the list into new rows → Click Expand button → Expand to New Rows.



Step 3: Convert to Date Format

1. Change the list into **Date format**.
2. Rename the column to "Leave Date".

Step 4: Group By → Count Employees Per Date

1. Select **Leave Date** column → Go to **Home Tab** → **Group By**.
 - Group By → **Basic**
 - New Column Name: **Days Count**
 - Operation: **Count Rows** → OK

The screenshot shows the Power BI Desktop interface. The 'Home' tab is active. The 'Group By' button in the 'Transform' ribbon is highlighted with a red box and a red circle with the number 2. A 'Group By' dialog box is open, showing the 'Date' column selected for grouping. The 'New column name' is 'Days Count' and the 'Operation' is 'Count Rows'. The 'Basic' radio button is selected. The 'OK' button is highlighted with a blue pen.

2. Result: Each date shows the **number of employees on leave**.

	Date	Days Count
	Valid 100%	Valid 100%
	Error 0%	Error 0%
	Empty 0%	Empty 0%
1	01-01-2025	2
2	02-01-2025	2
3	03-01-2025	3
4	04-01-2025	3
5	05-01-2025	2
6	06-01-2025	1
7	13-01-2025	2
8	12-01-2025	1
9	14-01-2025	1
10	15-01-2025	1
11	20-01-2025	2
12	19-01-2025	1
13	23-01-2025	1

Step 5: Group By → Count Employees Per Date

1. Select **Leave Date** column → Go to **Home Tab** → **Group By**.

- Group By → **Advanced**
 - New Column Name: **Days Count**
 - Operation: **Count Rows**
- **Add aggregation**
 - **New Column Name: List of Employees**
 - **Operations All Rows**
- **OK**

Group By

Specify the columns to group by and one or more outputs.

☐ Basic ☒ Advanced

Date

Add grouping

New column name

Days Count

List of Employees

Add aggregation

Operation

Count Rows

All Rows

Column

OK Cancel

Step 6: Combine Employee Names into Comma-Separated List

1. Add **Custom Column** → Formula:
2. **Text.Combine([List of Employees][Emp_name], ", ")**

Custom Column

Add a column that is computed from the other columns.

New column name
List of All Employees

Custom column formula ⓘ
= Text.Combine([List of Employees][Emp_name], ", ")

Available columns
Date
Days Count
List of Employees

< < Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

Cancel

- Combines employee names with comma and space separator.
- Delete the original list column if not needed.
- The final table looks like:

	Date	Days Count	List of Employees	List of All Employees
	Valid100% Error0% Empty0%	Valid100% Error0% Empty0%	Valid100% Error0% Empty0%	Valid100% Error0% Empty0%
1	01-01-2025	2	Table	Ajit Sharma, Vijay Singh
2	02-01-2025	2	Table	Ajit Sharma, Vijay Singh
3	03-01-2025	3	Table	Ajit Sharma, Vijay Singh, Vishal Sawant
4	04-01-2025	3	Table	Ajit Sharma, Vijay Singh, Vishal Sawant
5	05-01-2025	2	Table	Vijay Singh, Vishal Sawant
6	06-01-2025	1	Table	Vijay Singh
7	13-01-2025	2	Table	Rani Maurya, Ajay Kale
8	12-01-2025	1	Table	Ajay Kale
9	14-01-2025	1	Table	Ajay Kale
10	15-01-2025	1	Table	Ajay Kale
11	20-01-2025	2	Table	Sejal Singh, Arti Pandey
12	19-01-2025	1	Table	Arti Pandey
13	23-01-2025	1	Table	Vinay Jadhav

Step 7: Load Data Back to Excel

- Home → Close & Load To → choose **Table** in an existing or new worksheet.
- Your table is now dynamic:
 - Any new leave data added → just click **Refresh** to update results automatically.

Chapter 7: Power Query: Dynamic Removal of Junk Rows & Appending Data

Introduction

This is an advanced chapter where we will learn how to clean datasets with junk rows (blank or unnecessary rows at the top) dynamically and append multiple files using Power Query.

Sometimes, when downloading data (e.g., CSV or Excel files), there are extra rows at the top which are irrelevant. These rows vary from file to file:

- **Example:**
 - Central file: 3 junk rows → actual data starts from row 4
 - South file: 5 junk rows → actual data starts from row 6
 - West file: 2 junk rows → actual data starts from row 3

We need a method to dynamically remove these junk rows, regardless of how many there are in each file.

Step 1: Get Data from Folder

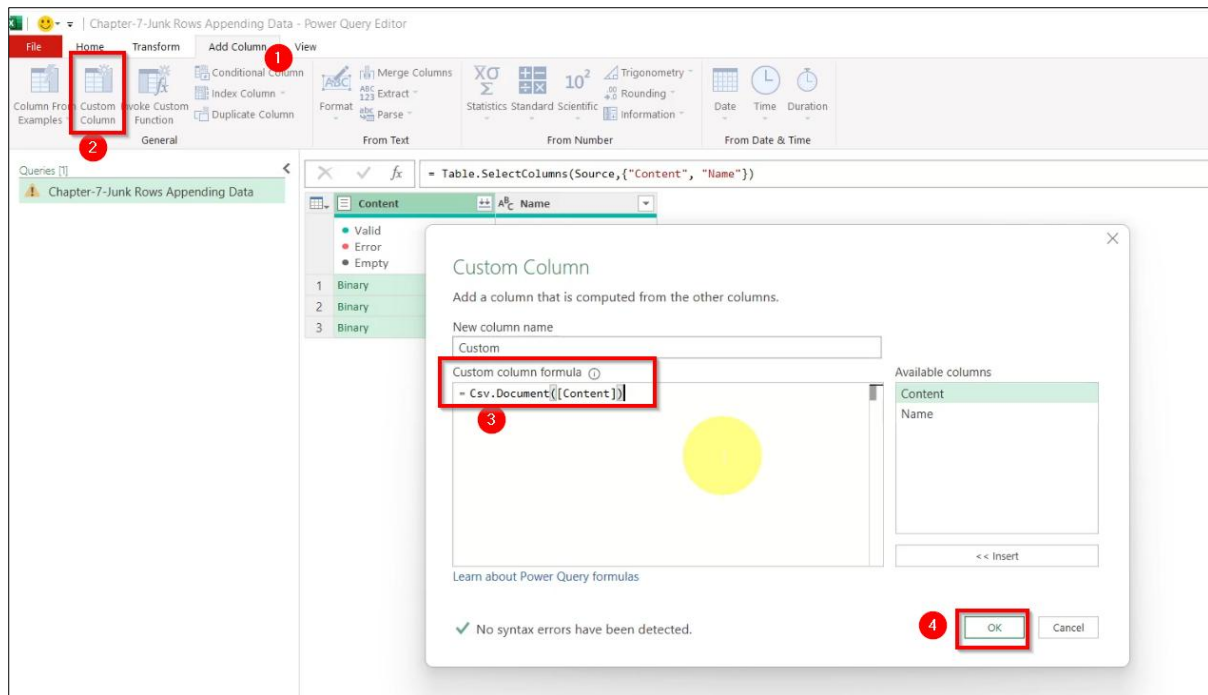
1. Go to **Data → Get Data → From File → From Folder**.
2. Select the folder containing all your files (CSV or Excel).
3. Click **Open**.
4. In the Navigator, click **Transform Data** to open the Power Query Editor.

Step 2: Keep Only Required Columns

1. Identify the columns you need (for example: **Content** and **Name**).
2. Select these columns while holding CTRL Key.
3. **Right-click → Remove Other Columns** to delete all unnecessary columns.

Step 3: Extract Data from Content Column

1. Click **Add Column Tab → Custom Column**.
2. Use the formula to extract the content: **Csv.Document([Content])**
 - Ensure Csv is capitalized correctly.
 - Double-click the column name when writing the formula.
3. Click **OK** to create the new table column.

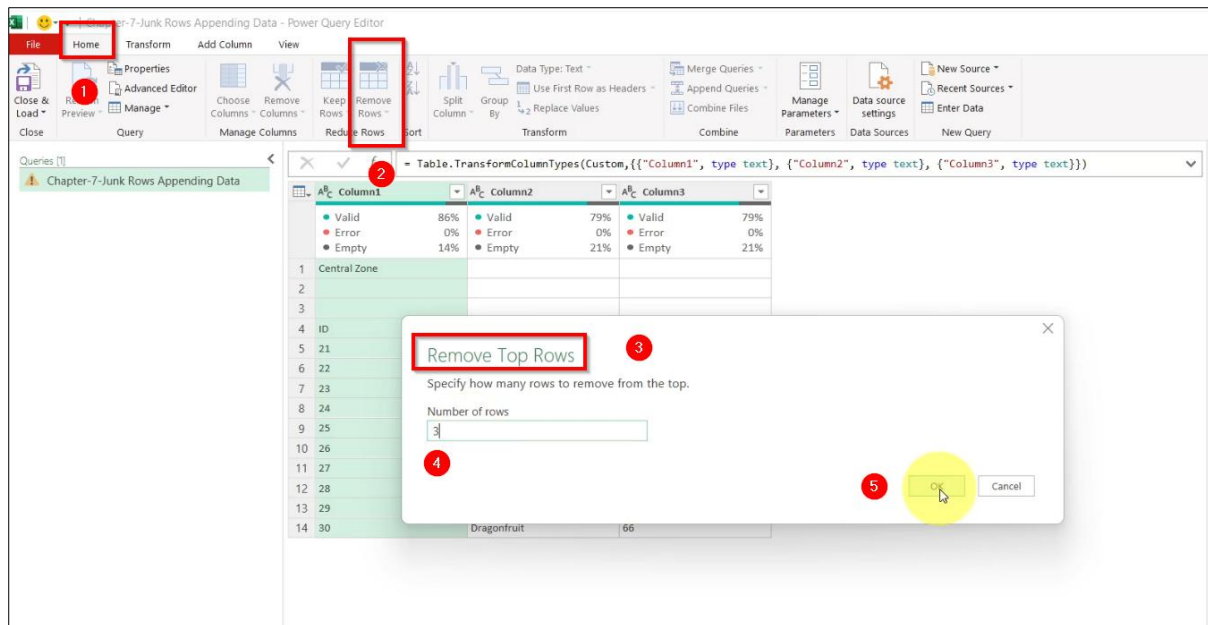


Step 4: Understanding Junk Rows

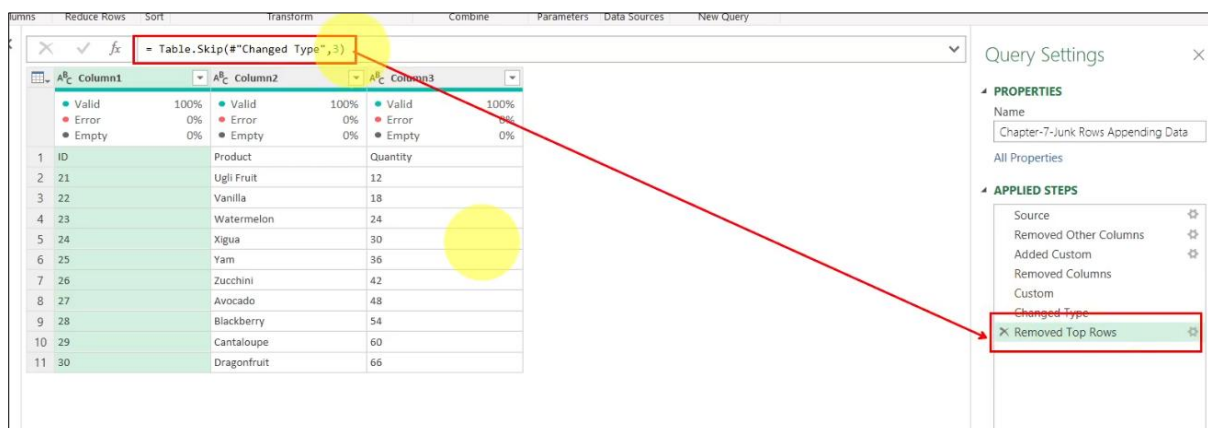
1. Preview the data: different files have different numbers of top junk rows.
2. We need a formula that can dynamically identify and skip these rows.
3. Temporary steps:
 - Expand tables for preview.
 - Remove binary columns if unnecessary.
 - Identify the number of rows to skip dynamically.

Step 5: Remove Top Rows (Dynamic Approach)

1. Initially, remove top rows manually:
 - **Home → Remove Rows → Remove Top Rows**
 - Enter a static number (e.g., 3 for Central file)

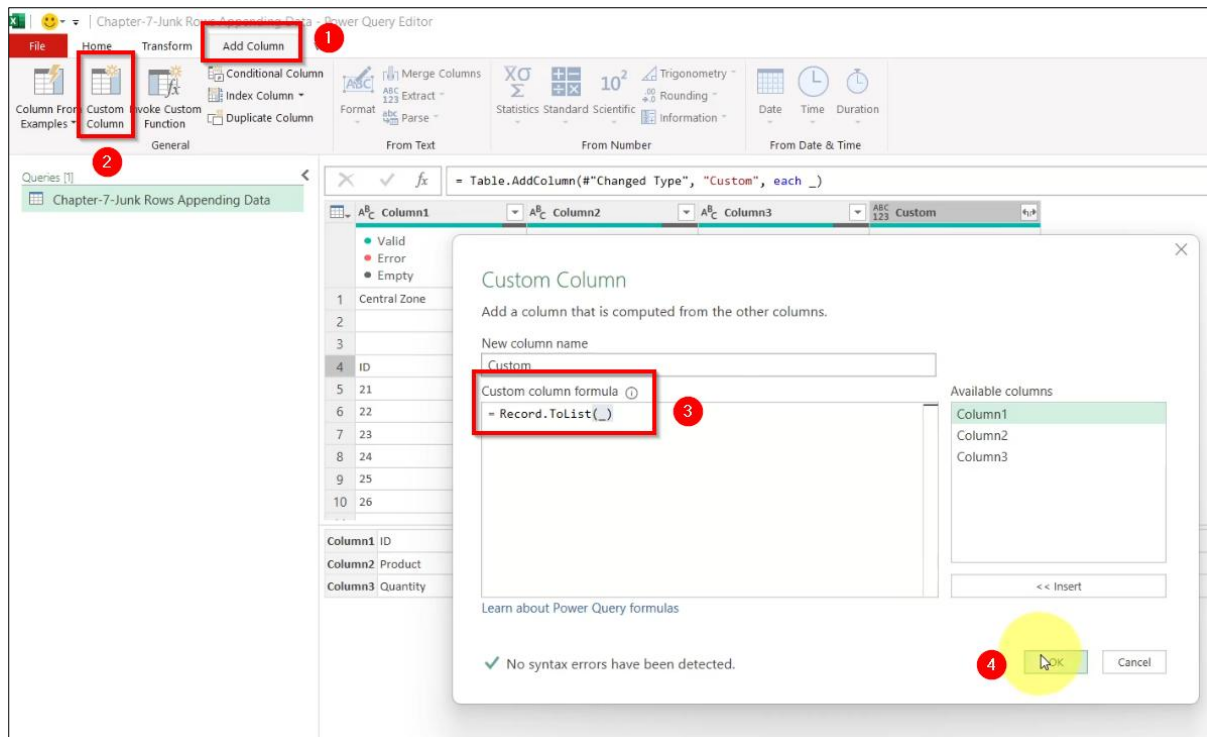


2. Observe the generated M formula in Power Query: **Table.Skip(PreviousStep, 3)**



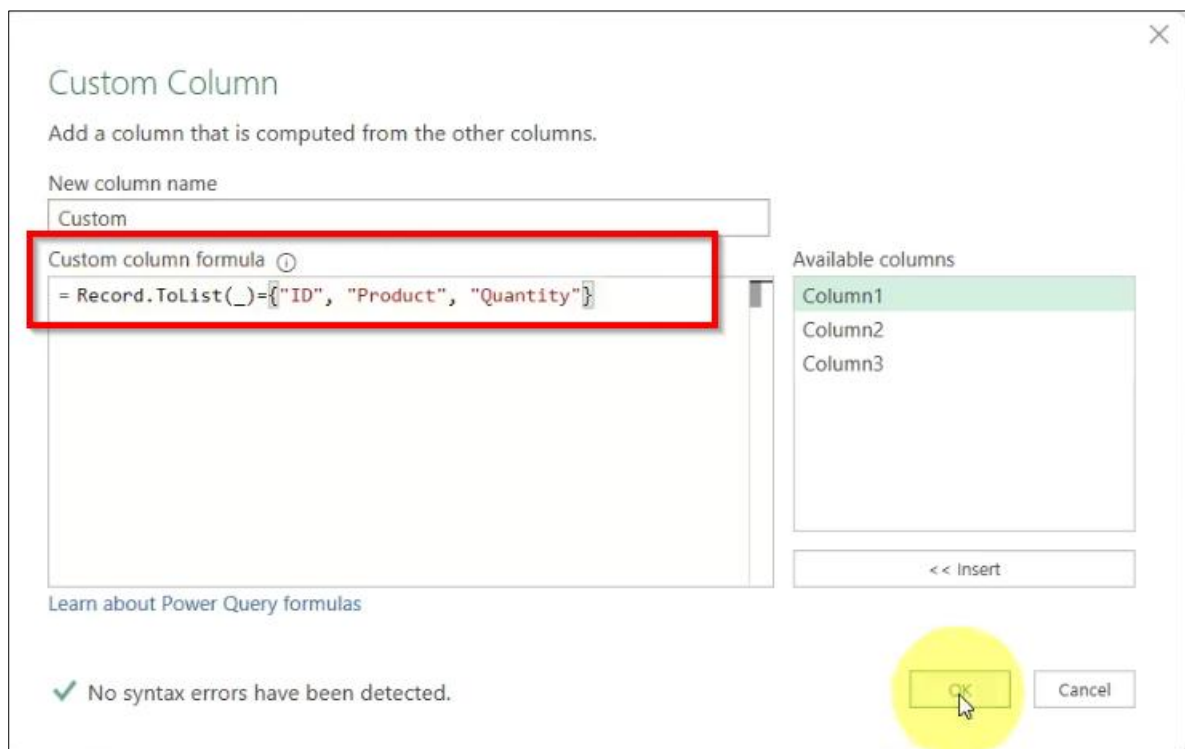
3. This is static. To make it dynamic:

- Create a Custom Column with _ as placeholder.
- Convert records to lists: Record.ToList(_)
- This prepares the row for dynamic evaluation.



Step 6: Identify Header Row

1. Determine which row contains the actual headers (ID, Product, Quantity).
2. **Add a Column > Custom Column**
3. Use a list comparison formula to find the row:
4. **Record.ToList(_)={"ID", "Product", "Quantity"}**



5. This returns True for the row with the header and False for all other rows.

	Column1	Column2	Column3	Custom
	Valid 86%	Valid 79%	Valid 79%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 14%	Empty 21%	Empty 21%	Empty 0%
1	Central Zone			FALSE
2				FALSE
3				FALSE
4	ID	Product	Quantity	TRUE
5	21	Ugli Fruit	12	FALSE
6	22	Vanilla	18	FALSE
7	23	Watermelon	24	FALSE
8	24	Xigua	30	FALSE
9	25	Yam	36	FALSE
10	26	Zucchini	42	FALSE
11	27	Avocado	48	FALSE
12	28	Blackberry	54	FALSE
13	29	Cantaloupe	60	FALSE
14	30	Dragonfruit	66	FALSE

Step 7: Reverse True/False for Dynamic Skipping

- Since Table.Skip removes rows until it encounters False, reverse logic:
 - Change Equal to Not Equal in the formula.
- Record.ToList(_)<>{"ID", "Product", "Quantity"}**

Custom Column

Add a column that is computed from the other columns.

New column name

Custom

Custom column formula

= Record.ToList(_)<>{"ID", "Product", "Quantity"}

Available columns

Column1
Column2
Column3

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

- This returns FALSE for the row with the header and TRUE for all other rows.

	Column1	Column2	Column3	Custom
	Valid 86% Error 0% Empty 14%	Valid 79% Error 0% Empty 21%	Valid 79% Error 0% Empty 21%	Valid 100% Error 0% Empty 0%
1	Central Zone			TRUE
2				TRUE
3				TRUE
4	ID	Product	Quantity	FALSE
5	21	Ugli Fruit	12	TRUE
6	22	Vanilla	18	TRUE
7	23	Watermelon	24	TRUE
8	24	Xigua	30	TRUE
9	25	Yam	36	TRUE
10	26	Zucchini	42	TRUE
11	27	Avocado	48	TRUE
12	28	Blackberry	54	TRUE
13	29	Cantaloupe	60	TRUE
14	30	Dragonfruit	66	TRUE

Step 8: Add Each and Promote Header

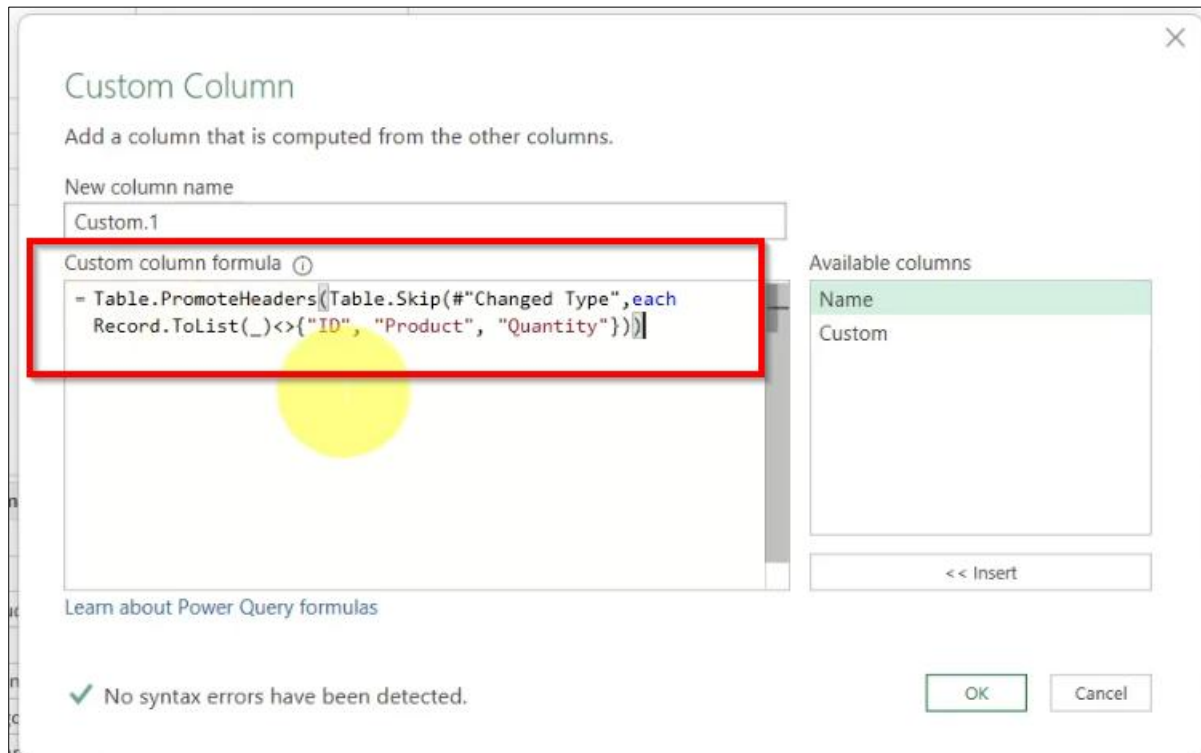
1. Now Copy the formula and remove the step
2. Home > Remove Top Rows > 3 > Ok
3. Edit the formula to include each at the beginning:
4. **Table.Skip(PreviousStep, each Record.ToList() <> {"ID", "Product", "Quantity"})**
5. After dynamically skipping junk rows, promote the header:
6. **Table.PromoteHeaders(Table.Skip(PreviousStep, each Record.ToList() <> {"ID", "Product", "Quantity"}))**
 - o Add a space before Table.PromoteHeaders if necessary to avoid dropdown conflicts in Power Query.

= Table.PromoteHeaders(Table.Skip(#"Changed Type",each Record.ToList()<>{"ID", "Product", "Quantity"}))			
ID	Product	Quantity	
21	Ugli Fruit	12	
22	Vanilla	18	
23	Watermelon	24	
24	Xigua	30	
25	Yam	36	
26	Zucchini	42	
27	Avocado	48	
28	Blackberry	54	
29	Cantaloupe	60	
30	Dragonfruit	66	

Step 9: Apply Formula to All Files

1. Now remove the all the previous steps to **Remove Column**

2. **Add a Column > Custom Column**
3. Copy this formula and apply it to all tables in the folder.



4. The formula dynamically detects the number of junk rows in each file.
5. No need to manually remove top rows for each file.

Note: Make sure to update the last steps to be replace with #changed Type or previous step:

Step 10: Final Result

1. All top junk rows are removed dynamically.
2. Headers are correctly identified and promoted.
3. Multiple tables can now be appended into one consolidated dataset.

✓ Outcome:

You can now clean and append multiple CSV/Excel files, no matter how many junk rows they contain, using dynamic Power Query formulas.

Chapter 8: Power Query: Real Job Data Transformation & Append

Objective

- Combine multiple CSV files (170–180 files in the example) into one clean table.
- Extract **top 7 rows** (header info) and **last row** (final values) from each file.
- Transform headers to vertical, remove unnecessary rows, and keep final format ready for Excel.

	C	D	E	F	G	H	I	J	K	L	M
1	Alpha Trade Corp	Omega Distribution	Nova Traders Ltd	Titan Power Group	Sparkline Pvt Ltd	EchoGrid Solutions	Zenith Networks	Quantum Flux Ltd	BlazeTrade Inc	GridLogic Partners	NeoSpark Venture
2	West Zone	East Sector	Central Block	North Sector	South Division	West Region	East District	North Wing	Central Node	South Corridor	West Line
3	WZ Supply Hub	ES Renewable Center	CB Energy Unit	NS Clean Supply	SD Power Plant	WR Energy Base	ED Grid Centre	NW Green Flow	CN Gen Systems	SC Utility Station	WL Hybrid Source
4	North Zone	South Sector	West Zone	East Sector	North Division	Central Region	West District	South Wing	East Node	North Corridor	Central Line
5	NZ Power Grid	SS Thermal Exchange	WZ Electric Coop	ES Utility Services	ND Green Grid	CR Smart Utility	WD CleanTech Services	SW Connect Ltd	EN Distribution Ltd	NC Power Authority	CL Renewable Net
6	http://docs.fake/1001	http://docs.fake/1002	http://docs.fake/1003	http://docs.fake/1004	http://docs.fake/1005	http://docs.fake/1006	http://docs.fake/1007	http://docs.fake/1008	http://docs.fake/1009	http://docs.fake/1010	http://docs.fake/1011
7	APL-X1001	APL-X1002	APL-X1003	APL-X1004	APL-X1005	APL-X1006	APL-X1007	APL-X1008	APL-X1009	APL-X1010	APL-X1011
8	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
9											
10		0	-20	-46	0	-100	-100	-200	0	-37.76	-69.04
11		0	-20	-46	0	-100	-100	-200	0	-37.76	-69.04
12		0	-20	-46	0	-100	-100	-200	0	-37.76	-69.04
13		0	-20	-46	0	-100	-100	-200	0	-37.76	-69.04
14		0	-20	-46	0	-100	-100	-200	0	-37.76	-69.04
15		0	-20	-46	0	-100	-100	-200	0	-37.76	-69.04
16		0	-20	-46	0	-100	-100	-200	0	-37.76	-69.04
17		0	-20	-46	0	-100	-100	-200	0	-37.76	-69.04
18		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04
19		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04
20		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04
21		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04
22		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04
23		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04
24		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04
25		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04
26		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04
27		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04
28		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04
29		0	-20	-46	0	-100	-200	-200	0	-37.76	-69.04

Step 1: Load Data from Folder

1. Go to **Data → Get Data → From Folder**.
2. Select folder containing all CSV files.
3. Click **Transform Data** to open Power Query.
4. Remove unnecessary columns (like Binary content) to simplify table view.

Step 2: Extract Top Rows (Header)

1. Click the **table column** to expand data.
2. Go to **Home → Keep Rows → Keep Top Rows**.
3. Enter **7** to keep top 7 rows.
4. This becomes your **header section**.
5. Copy the M formula (Table.FirstN) to Notepad for reuse later.

Step 3: Extract Bottom Row (Final)

1. Undo previous step to revert table.
2. Go to **Home → Keep Rows → Keep Bottom Rows**.
3. Enter **1** to keep last row (final row).
4. Copy M formula (Table.LastN) to Notepad.

Step 4: Combine Top & Bottom Rows

1. Use **Add Column** → **Custom Column**.
2. Paste formulas for top and bottom rows using & (join).
3. This creates a **single column with both header & final row**.

Custom Column

Add a column that is computed from the other columns.

New column name

Custom

Custom column formula ⓘ

```
= Table.ReplaceValue([Custom.1], "", "Final",  
Replacer.ReplaceValue, {"Column1"})
```

Available columns

Name

Custom.1

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

Step 5: Transform / Transpose Table

1. Expand the combined column to see data in rows.
2. Go to **Transform** → **Transpose** to make headers vertical.
3. Remove unnecessary columns (e.g., temporary custom columns).

Step 6: Promote Headers

1. After transpose, go to **Home** → **Use First Row as Headers**.
2. Now your **columns have proper names** (Seller Name, Source, Destination, etc.).

Step 7: Clean Final Table

- Remove blank rows or extra totals.
- Check all tables for consistency.
- Load the final table into Excel:
 - **Close & Load To** → **Excel** → **Table Format** → **New Worksheet** (or existing sheet).

Step 8: Refresh for New Files

- Add new CSV files to the folder.
- In Excel, simply **Refresh** Power Query.
- All steps (Top 7 rows + Last row + Transpose + Headers) are applied automatically.

Chapter 9: Use of Delimiters (Power Query)

What is a Delimiter in Power Query?

A delimiter is a symbol, character, or space used to separate values inside one cell.

Example

If a cell has: **Laptop, Monitor, Mouse** here comma (,) is the delimiter → it separates the products.

Common Delimiters

- Comma (,) → Apple, Mango, Banana
- Semicolon (;) → A;B;C
- Space () → First Last
- Tab → often used in text files.
- Custom delimiters → You can set any character (like | or =).

In Power Query

- Power Query uses delimiters to split data into separate parts.
- You can Split Column by Delimiter → choose comma, space, tab, or custom.
- You can split into:
 - Columns → products in separate columns.
 - Rows → products listed down in multiple rows (more useful for analysis).

Problem

- We have a file with **Customer ID** and **Product List**.
- Products are written together, separated by a **comma (,)**.
Example: Laptop, Monitor, Mouse

Customer ID	Product List
1001	Laptop, Monitor, Keyboard,
1002	Mouse, External Hard Drive,
1003	Laptop, Mouse, Keyboard, Monitor
1004	Solid State Drive (SSD), Graphics Card (GPU)
1005	Motherboard, Processor (CPU), Mouse
1006	Cooling Fan, Webcam, Printer, Scanner,
1007	Router, USB Flash Drive, Headphones, Laptop
1008	Laptop, Mouse, Keyboard, Monitor, Headphones
1009	Router, USB Flash Drive, Headphones, Laptop
1010	Keybaord, USB Flash Drive, Headphones, Laptop

- We need to:
 1. Get a **unique list of products**.
 2. Find **how many times each product was ordered**.

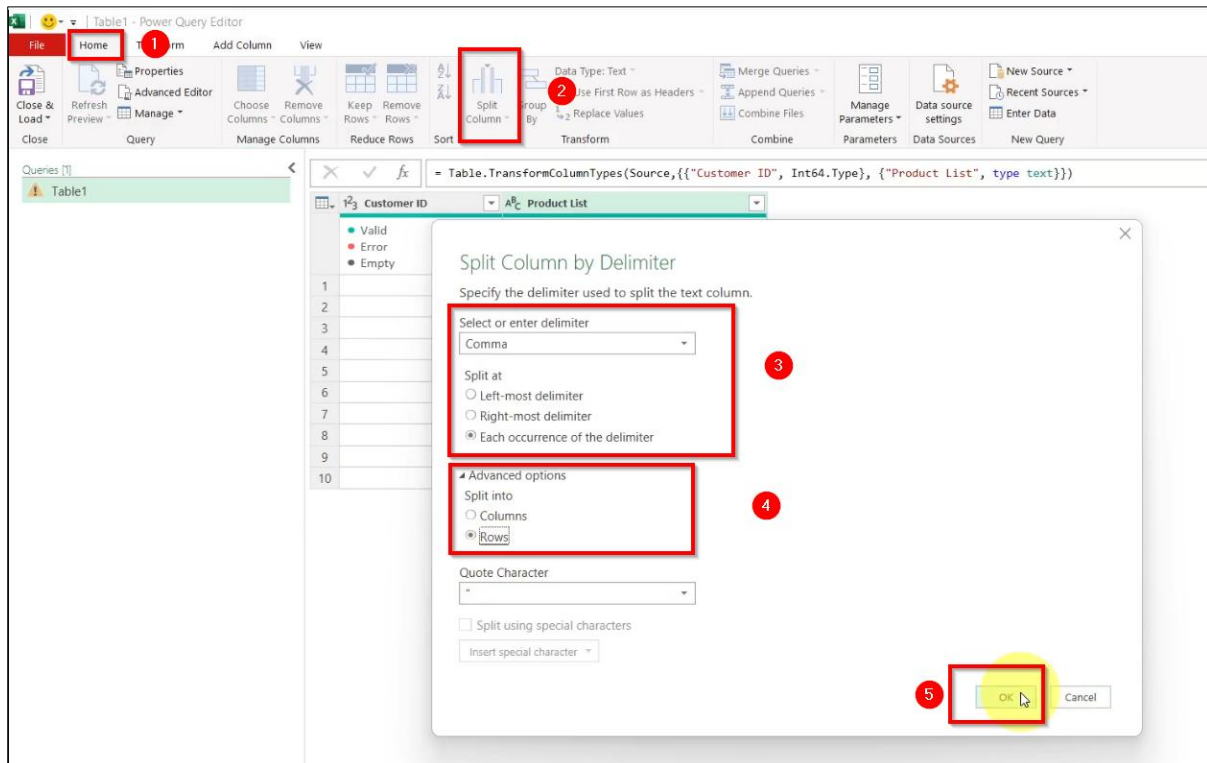
Solution Steps

1. Load Data

- Open the file in Power Query. **Data > From Table/Range**
- You will see Customer ID and Product List.

2. Split by Delimiter

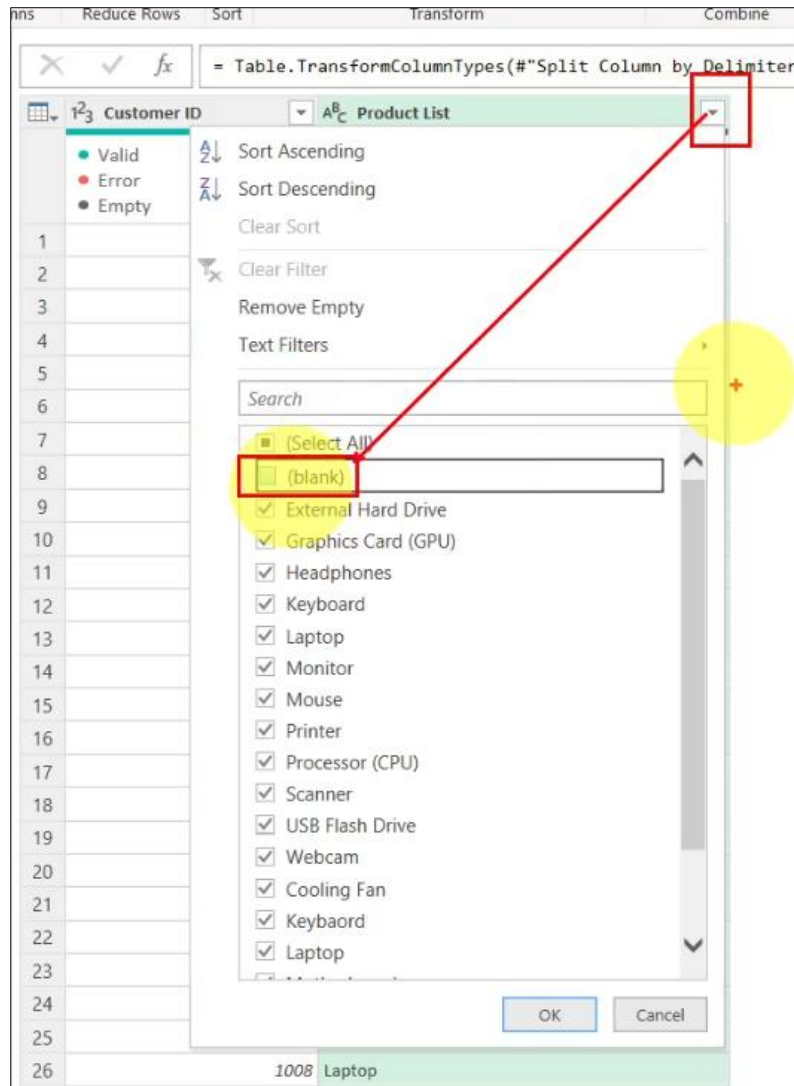
- Select the Product List column.
- Use **Split Column → By Delimiter (Comma)**.
- In options, choose **Split into Rows** (not columns).



- Now each product comes in a separate row.

3. Remove Empty Rows

- After splitting, some rows may be blank.
- Filter and remove blank values.



4. Clean Data

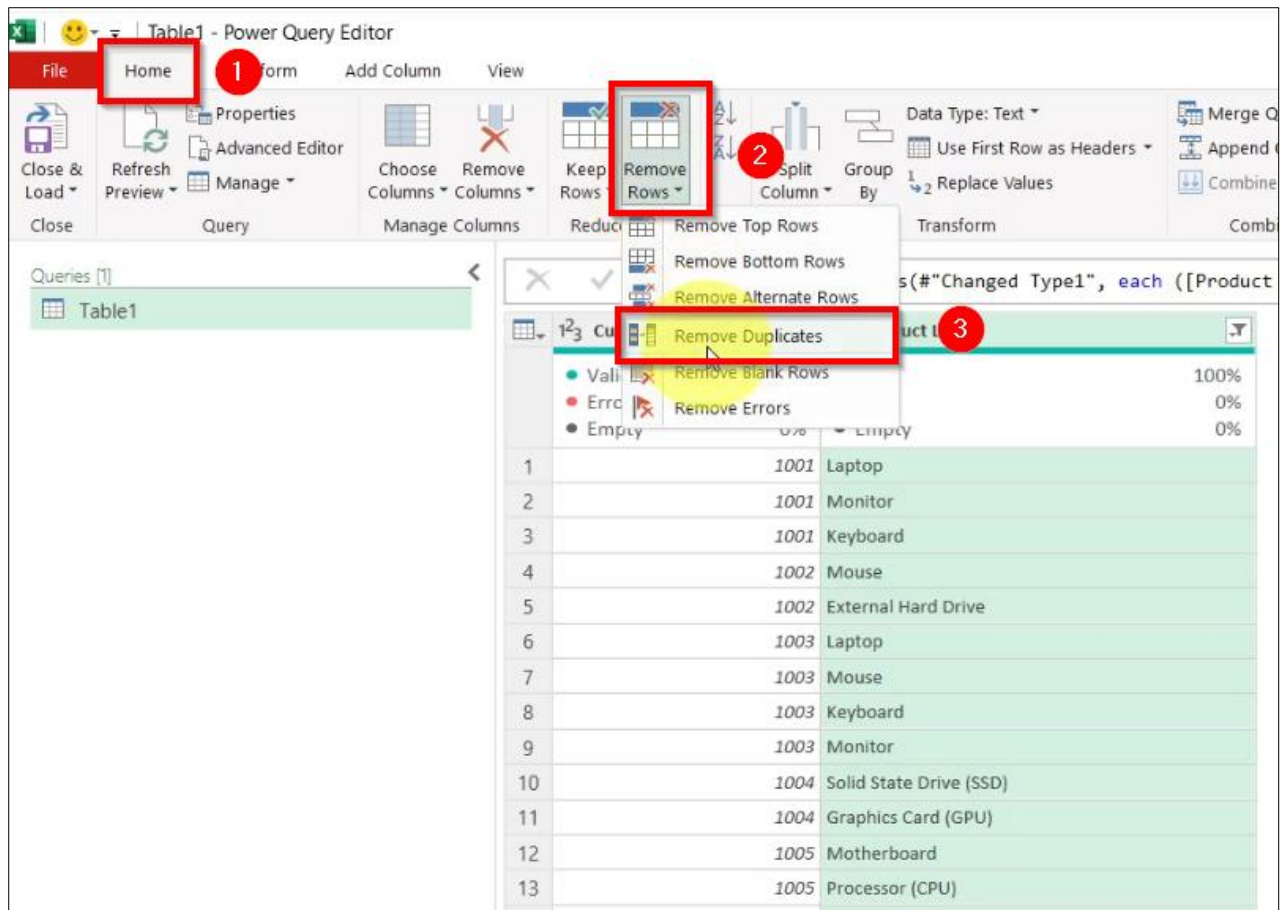
- Sometimes products have **extra spaces** or **spelling mistakes**.
- Use **Trim** to remove spaces.
- Use **Replace Values** to fix spelling errors.

The screenshot shows a Power BI Desktop interface with a data table. The table has three columns: Customer ID, Product List, and a third column (likely a measure or calculated column). The Product List column is highlighted, and a context menu is open over it. The menu includes options like Copy, Remove, Duplicate Column, Add Column From Examples..., Remove Duplicates, Remove Errors, Change Type, Transform, Replace Values..., Replace Errors..., Create Data Type, Split Column, Group By..., Fill, Unpivot Columns, Unpivot Other Columns, Unpivot Only Selected Columns, Rename..., Move, Drill Down, and Add as New Query. The Transform option is selected, and its submenu is open, showing options like lowercase, UPPERCASE, Capitalize Each Word, Trim, and Clean. The Trim option is highlighted.

Customer ID	Product List
1	Laptop
2	Monitor
3	Keyboard
4	Mouse
5	External Hard Drive
6	Laptop
7	Mouse
8	Keyboard
9	Monitor
10	Solid State Drive (SSD)
11	Graphics Card (GPU)
12	Motherboard
13	Processor (CPU)
14	Mouse
15	Cooling Fan
16	Webcam
17	Printer
18	Scanner
19	Router
20	USB Flash Drive
21	Headphones
22	Laptop
23	Laptop

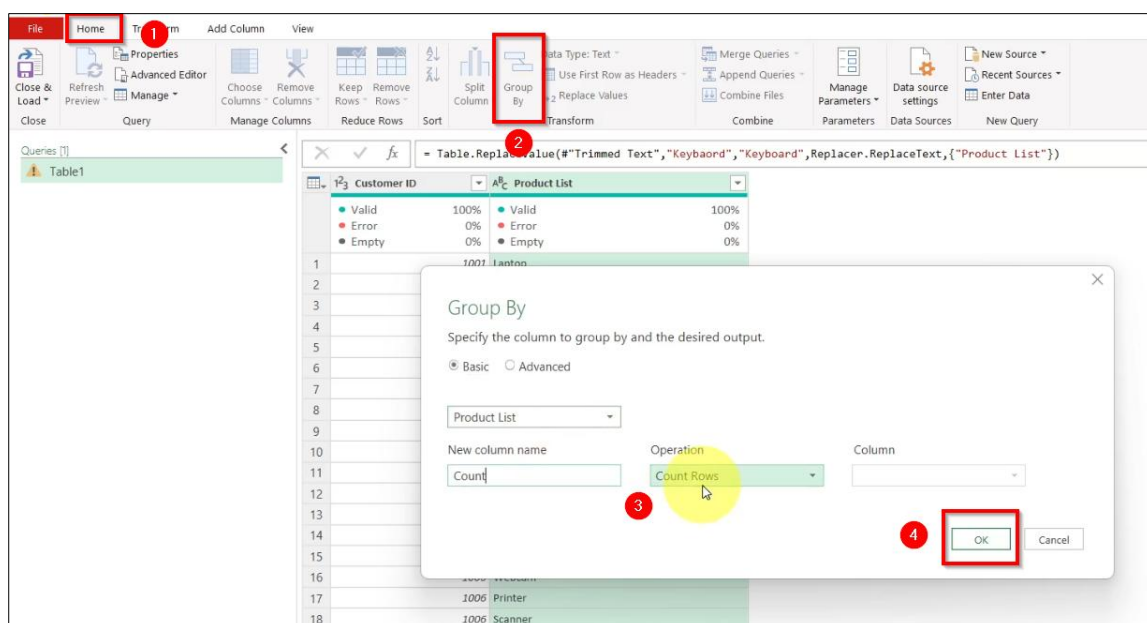
5. Unique Product List

- Go to **Remove Duplicates**.
- Now you get a clean list of unique products.



6. Count Products

- If you want to know how many times each product is ordered:
 - Don't remove duplicates.
 - Instead, use **Group By** → **Product List** → **Count Rows**.



Chapter 10: Real Job Query (Power Query)

Objective

- Clean messy data fetched from a website.
- Software name and multiple versions are in a single cell.
- Convert this into a proper structured table.
- Find how many versions each software has.

Problem

- Data was **fetched from a website**.
- Each row has **Software name** and **all its versions** written in one cell.

Example:

- Microsoft Windows 95, 98, XP, 7, 8, 10
- Goal:
 1. Get all software versions in a **clean column format (unpivoted)**.
 2. Count **how many versions each software has**.

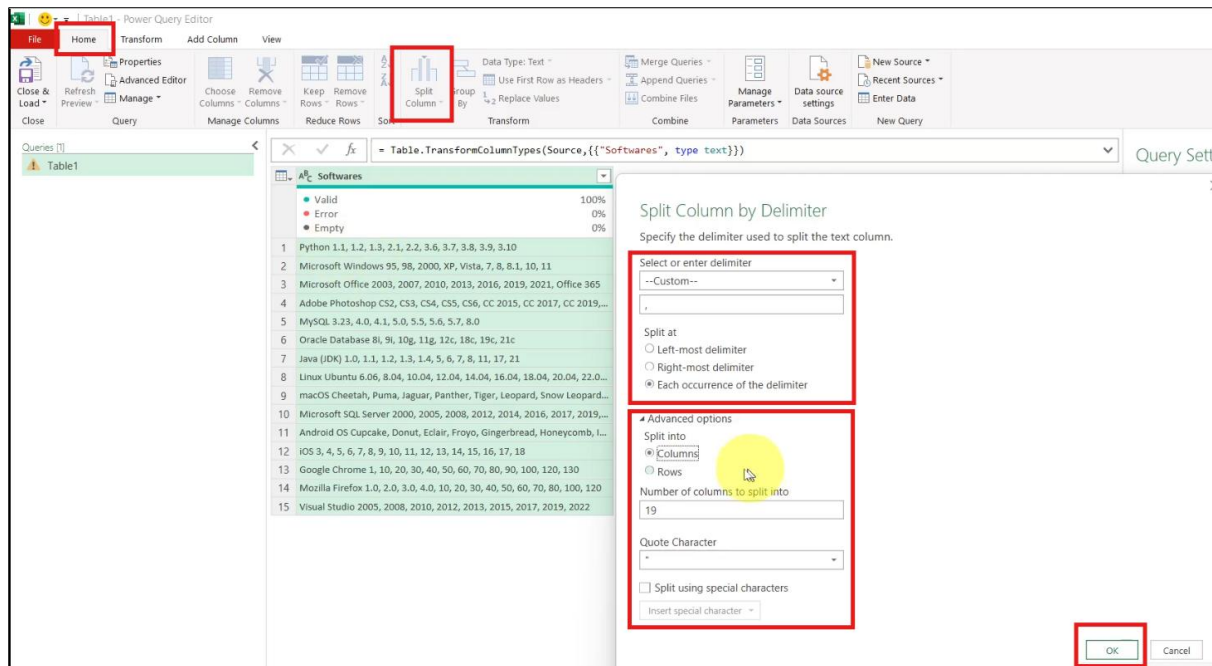
	A
1	Softwares
2	Python 1.1, 1.2, 1.3, 2.1, 2.2, 3.6, 3.7, 3.8, 3.9, 3.10
3	Microsoft Windows 95, 98, 2000, XP, Vista, 7, 8, 8.1, 10, 11
4	Microsoft Office 2003, 2007, 2010, 2013, 2016, 2019, 2021, Office 365
5	Adobe Photoshop CS2, CS3, CS4, CS5, CS6, CC 2015, CC 2017, CC 2019, CC 2021, CC 2023
6	MySQL 3.23, 4.0, 4.1, 5.0, 5.5, 5.6, 5.7, 8.0
7	Oracle Database 8i, 9i, 10g, 11g, 12c, 18c, 19c, 21c
8	Java (JDK) 1.0, 1.1, 1.2, 1.3, 1.4, 5, 6, 7, 8, 11, 17, 21
9	Linux Ubuntu 6.06, 8.04, 10.04, 12.04, 14.04, 16.04, 18.04, 20.04, 22.04, 24.04
10	macOS Cheetah, Puma, Jaguar, Panther, Tiger, Leopard, Snow Leopard, Lion, Mavericks, Catalina, Big Sur, Monterey, Ventura, Sonoma
11	Microsoft SQL Server 2000, 2005, 2008, 2012, 2014, 2016, 2017, 2019, 2022
12	Android OS Cupcake, Donut, Eclair, Froyo, Gingerbread, Honeycomb, Ice Cream Sandwich, Jelly Bean, KitKat, Lollipop, Marshmallow, Nougat, Oreo, Pie, 10, 11
13	iOS 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18
14	Google Chrome 1, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 120, 130
15	Mozilla Firefox 1.0, 2.0, 3.0, 4.0, 10, 20, 30, 40, 50, 60, 70, 80, 100, 120
16	Visual Studio 2005, 2008, 2010, 2012, 2013, 2015, 2017, 2019, 2022
17	
18	

Step 1: Load Data

- Load Data using Data > From Table/Range
- Convert Range to Table (Tick My Table Has Headers)
- Import table into Power Query.
- If headers are wrong → use **Use First Row as Headers**.

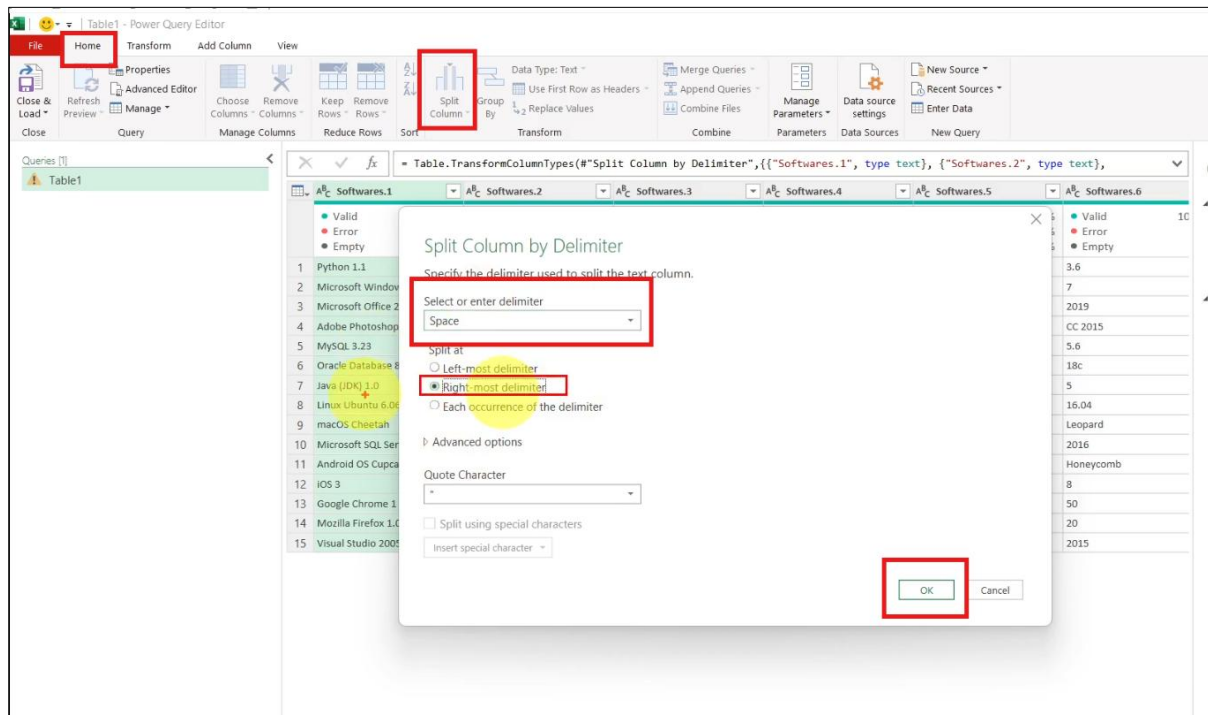
Step 2: Split by Delimiter (Comma + Space)

- Select the **Versions column**.
- Choose **Split Column → By Delimiter**.
- Select Custom and use **Comma + Space (,)**.
- Now each version will appear in a separate column.



Step 3: Split Software Name & First Version

- Some rows still have **Software name + first version** together.
- Use **Split Column → By Delimiter → Right-most Space**.
- This separates the **Software Name** from the version.



Step 4: Unpivot Columns

- Select the **Software Name** column.
- Then use **Unpivot Other Columns**.
- All versions will now appear **in one column under each software**.

Table1 - Power Query Editor

Transform

Unpivot Other Columns

Table1

	Software.1.1	Software.1.2	Software.2	Software.3	Software.4	Software.5
1	Python	1.1	1.2	1.3	2.1	2.2
2	Microsoft Windows	95	98	2000	XP	Vista
3	Microsoft Office	2003	2007	2010	2013	2016
4	Adobe Photoshop	CS2	CS3	CS4	CS5	CS6
5	MySQL	3.23	4.0	4.1	5.0	5.5
6	Oracle Database	8i	9i	10g	11g	12c
7	Java (JDK)	1.0	1.1	1.2	1.3	1.4
8	Linux Ubuntu	6.06	8.04	10.04	12.04	14.04
9	macOS	Cheetah	Puma	Jaguar	Panther	Tiger
10	Microsoft SQL Server	2000	2005	2008	2012	2014
11	Android OS	Cupcake	Donut	Eclair	Froyo	Gingerbread
12	iOS	3	4	5	6	7
13	Google Chrome	1	10	20	30	40
14	Mozilla Firefox	1.0	2.0	3.0	4.0	10
15	Visual Studio	2005	2008	2010	2012	2013

Step 5: Group & Count (Optional)

- Go to **Home** → **Group By**.
- Group by **Software Name**.
- Operation: **Count Rows**.
- This shows how many versions each software has.

Table1 - Power Query Editor

Home

Group By

Table1

Group By

Specify the column to group by and the desired output.

Basic

Software.1.1

New column name: Count

Operation: Count Rows

Column:

OK Cancel

✓ Final Output

- Data is now **clean and structured**.
- Each row has → **Software Name | Version**.
- Easy to filter, analyze, and count versions.

